

Multitrait GWAS Analysis of Brain MRI Phenotypes: Combinatorial challenge

Statistical Genetics unit at Institut Pasteur

Christopher Won supervised by Dr. Annaïg DeWalsche and Dr. Hugues Aschard

August, 2026

Contents

1	Introduction: Background	3
1.1	Statistical Genetics	3
1.2	Genotypic Data	3
1.3	Phenotypic Data	3
1.4	Genome-Wide Association Studies	4
1.5	From Univariate to Multitrait Analysis	4
2	Introduction: Project Overview and Motivation	5
2.1	The Combinatorial Challenge	5
2.2	Pipeline	6
3	Pilot Analysis: 5000 SNPs over 5 Phenotypes (UKB_5000x5.ipynb)	9
3.1	Dataset	9
3.2	Covariance Estimator: Naive Correlation	9
3.3	Results and Visualization	10
3.4	Limitations and Next Steps	10
4	Scaling Up: 100,000 SNPs over 10 Phenotypes (UKB_100kx10.ipynb)	10
4.1	Dataset	10
4.2	Covariance Estimator: JASS-inspired	11
4.3	Results and Visualization	12
4.4	Limitations and Next Steps	12
5	1,000,000 SNPs over 100 Phenotypes (UKB_1milx100.ipynb)	13
5.1	Dataset	13
5.2	Covariance Estimator: LDSC intercept matrix	13
5.3	Sampling Loop	14

5.4	Verification of the Cholesky Solver	14
5.5	Stability of the Sampling Method	15
5.6	Limitations and Next Steps	16
6	Halfway there: 5,000,000 SNPs over 500 Phenotypes	17
6.1	Input Dataset	17
6.2	Sampling $\mapsto$ Omnibus Test $\mapsto$ Rescaling	17
6.3	Verification of the Parallel Sampling Loop (<code>verify_sampling.py</code>)	18
6.4	Limitations and Next Steps	18
7	Fullscale: 9,611,261 SNPs by 1,010 Phenotypes	19
7.1	Input Dataset	19
7.2	Sampling $\mapsto$ Omnibus Test $\mapsto$ Rescaling	19
7.3	Verification of the Parallel Sampling Loop (<code>verify_sampling_fullscale.py</code>)	21
7.4	Post-Run Data Integrity Check	21
7.5	Limitations and Next Steps	21
8	Fullscale: Downstream Analysis	22
8.1	Phenotype Score Matrix Computation	22
8.2	Robustness of the Score Estimates	22
8.3	Comparing Univariate, Joint, and Subset-Based P-values	23
8.4	Phenotype Score vs. Univariate Signal	25
8.5	Characterizing Subset-Specific Discoveries	26
8.6	Cumulative Joint Testing of Score-Ranked Phenotypes	27
8.6.1	Comparing P-values: Univariate, Joint, and Optimal Subset	34
9	Conclusion and Next Steps	37
10	Appendix	41
10.1	Claim: LDSC intercept matrix is an unbiased estimator of the null covariance matrix	41
10.2	Detailed derivation of JASS-inspired null covariance matrix	50
10.3	Claim: Rescaling Transformation Preserves Tail Probabilities	51
10.4	Matrix Inversion via Cholesky factorization	53
	Glossary of Key Terms	54

1 Introduction: Background

We begin with a section that introduces the concepts needed to follow the rest of the report: statistical genetics, the genotypic and phenotypic data underlying this analysis, the standard GWAS testing procedure, and the distinction between univariate and multitrait analysis. These concepts are essential in understanding the motivation behind this paper as addressed in Section 2. Readers already familiar with GWAS may skip ahead; the **Glossary of Key Terms** remains available throughout as a quick reference for individual terms. All terms in **bold** can be found in the glossary for precise definitions.

1.1 Statistical Genetics

Statistical genetics is the branch of statistics concerned with relating genetic variation to observable traits, combining population genetics, probability theory, and large-scale computation to detect associations between **DNA** variants and **phenotypes** across a cohort. In this project, the traits of interest are not disease outcomes or agronomic measures but structural brain MRI phenotypes and the cohort is a subset of **UK Biobank (UKB)** participants [1]. Whatever the trait under study, the underlying task is the same: test each of millions of genetic variants individually for association with the phenotype, then correct for the number of tests performed across the **genome**. This report extends that task from single phenotypes to combinations of correlated phenotypes tested jointly (Section 1.5).

1.2 Genotypic Data

The **genotype** of an individual is the information encoded in their DNA, a sequence of base pairs formed from four nucleotides Adenine (A), Thymine (T), Guanine (G) and Cytosine (C). Most genetic variation between individuals takes the form of **Single Nucleotide Polymorphisms (SNPs)**: single positions in the genome at which individuals carry different bases, and these are the variants we are focusing on. This report analyzes approximately 9.6 million SNPs, genotyped in a UKB cohort of 32,947 individuals.

Each SNP has two possible bases, or **alleles**; one is designated the *effect allele*. Since humans carry two copies of each chromosome, an individual’s genotype at a given SNP is encoded as the number of copies of the effect allele they carry: 0, 1, or 2. Stacking this count across all N individuals and M SNPs gives the genotype matrix $\mathbf{X} \in \mathbb{R}^{N \times M}$ used internally by GWAS (Section 1.4 below); this report never operates on $\mathbf{X}$ directly, only on statistics derived from it. Nearby SNPs on the same chromosome tend to be co-inherited and therefore correlated which is a property called **linkage disequilibrium (LD)**. This becomes relevant later when estimating the null covariance matrix Σ_K via **LDSC** over other estimators addressed in the following sections.

1.3 Phenotypic Data

The phenotypes analysed in this report are 1,010 structural measurements derived from brain MRI scans of the same UKB participants. These fall broadly into three categories: cortical surface area (e.g. `Area_BA1_left`), cortical thickness (e.g. `Mean_thickness_S-orbital-lateral_right`), and subcortical or whole-brain volumes (e.g. `Volume_Hippocampus_right`, `Volume_CSF_whole-brain`). Unlike a single disease outcome, these 1,010 phenotypes are not independent measurements: they are all derived from the same scan of the same individual, and many are anatomically and developmentally related, so substantial correlation between phenotypes is expected and must be accounted for statistically.

1.4 Genome-Wide Association Studies

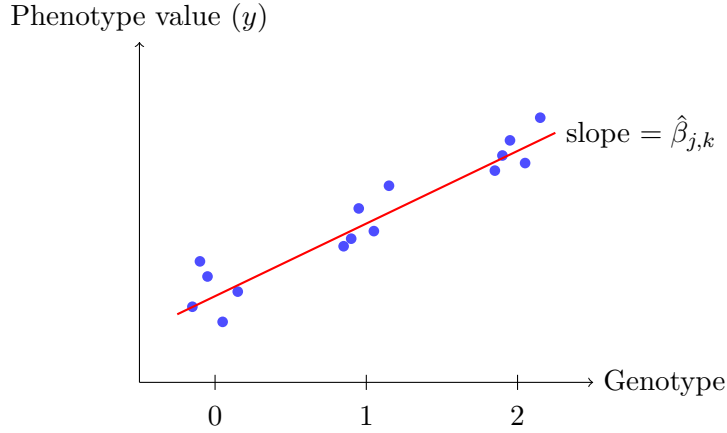


Figure 1: Conceptual illustration of a single SNP–phenotype regression: each point is one participant, plotted by genotype at SNP j and value for phenotype k , with slope $\hat{\beta}_{j,k}$ – the estimated change in phenotype k per additional copy of the effect allele at SNP j

A Genome-Wide Association Study (GWAS) [2] tests, SNP by SNP, whether carrying the effect allele is associated with a measured phenotype across the cohort. In our context for a single phenotype k , GWAS fits the linear model

$$y_k = \mathbf{X}\beta_k + \epsilon_k,$$

where $y_k \in \mathbb{R}^N$ is the phenotype vector, $\beta_k \in \mathbb{R}^M$ contains the effect size of each SNP on phenotype k , and ϵ_k is residual noise. For each SNP j , the test of interest is whether $\hat{\beta}_{j,k}$ differs significantly from zero, summarized by the **z-score** $z_{j,k} = \hat{\beta}_{j,k}/\text{SE}(\hat{\beta}_{j,k})$. Under the null hypothesis of no association: $\mu_{j,k} = 0$, $z_{j,k} \sim \mathcal{N}(0, 1)$; more generally, $z_{j,k} \sim \mathcal{N}(\mu_{j,k}/\text{SE}(\hat{\beta}_{j,k}), 1)$, where $\mu_{j,k}$ is the true effect size (derived in Appendix 10.2). GWAS has already been performed across all 1,010 phenotypes for this UKB cohort, producing a z-score matrix of shape approximately 9.6 million $\times$ 1,010; each entry is GWAS summary statistic. The regression above happens once, upstream, when the original GWAS is run; this report operates only on its output $\mathbf{X}$ (z-scores), never on individual-level phenotype or genotype data.

1.5 From Univariate to Multitrait Analysis

The GWAS model above is univariate: it tests one SNP against one phenotype at a time, independently of the other 1,009 phenotypes. Applied naively across all 1,010 phenotypes, this means running 1,010 separate tests per SNP, each blind to the fact that many of these phenotypes are correlated and may share a common underlying genetic signal. A SNP with a moderate, coordinated effect across many related brain structures may fail to reach significance in any single univariate test, yet be strongly supported when the phenotypes are considered jointly.

Multitrait analysis addresses this by testing the association between a SNP and several phenotypes simultaneously, accounting for their correlation, which can substantially increase statistical power for SNPs with such coordinated effects. The cost of this approach is combinatorial: with $K = 1,010$ phenotypes, there is no single, unique way to group phenotypes for a joint test. The number of possible groupings is intractably large. Section 2.1 formalizes this problem and Section 2.2 introduces the sampling-based strategy used throughout the rest of the report to address it.

2 Introduction: Project Overview and Motivation

This report documents the implementation and progressive validation of a random subset sampling strategy for multitrait GWAS analysis, applied to brain MRI phenotypes UKB. The work is part of a broader effort contributing to an ongoing paper called *‘Bypassing the combinatorial challenge in multitrait genetics studies’* from the Statistical Genetics unit at Institut Pasteur, which demonstrates that the otherwise intractable problem of selecting which phenotypes to test jointly can be resolved through random sampling. This is to determine which phenotype subsets carry the largest association signal against SNP(s). The specific aim here is to implement an existing pipeline, verify its correctness on tractable small-scale cases, and scale it progressively toward the full dataset of 1,010 phenotypes and approximately 9.6 million SNPs which requires optimization of the sampling methods.

Note: Each notebook in the progressive scaling pipeline loads a submatrix of increasing size, culminating in the full z-score matrix.

2.1 The Combinatorial Challenge

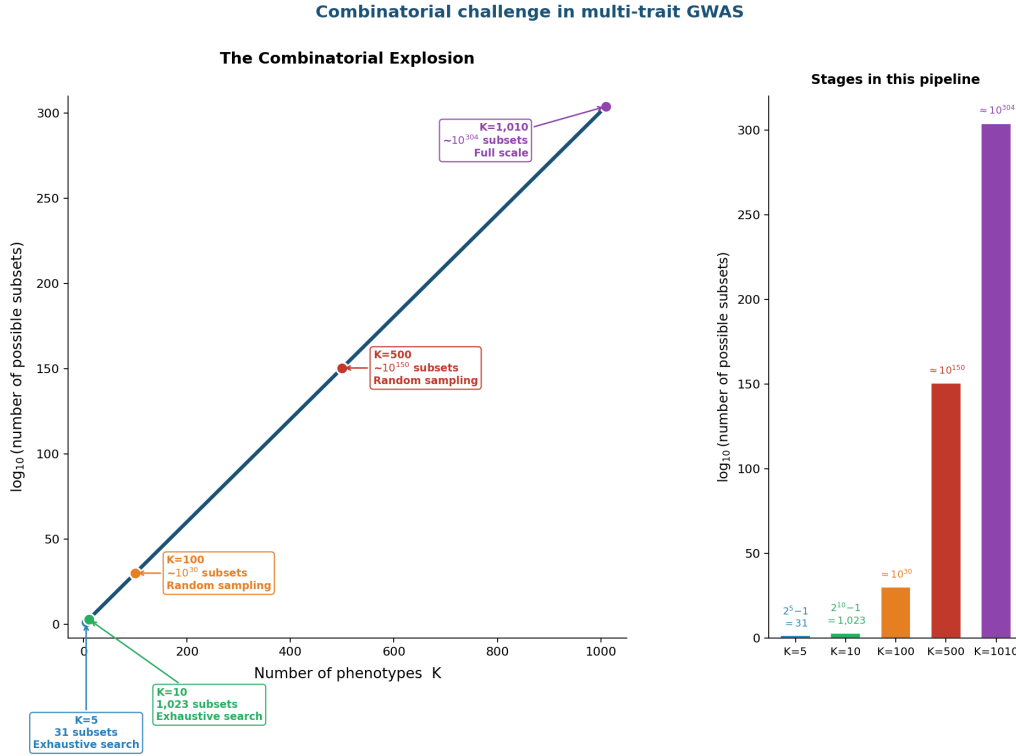


Figure 2

From a pool of K phenotypes, the number of possible phenotype subsets grows astronomically. For $K = 100$, there are over 10^{30} possible sets (Figure 2), making exhaustive search infeasible. Brain MRI phenotypes present a particularly challenging case: all 1,010 traits are derived from the same UK Biobank participants and are highly correlated, meaning the full covariance structure must be accounted for in every joint test via the omnibus statistic $S = \mathbf{z}^\top \hat{\Sigma}_K^{-1} \mathbf{z}$ where $\mathbf{z}$ is the vector of GWAS z-scores for a SNP across K selected phenotypes and $\hat{\Sigma}_K$ is their $K \times K$ LDSC intercept matrix.

2.2 Pipeline

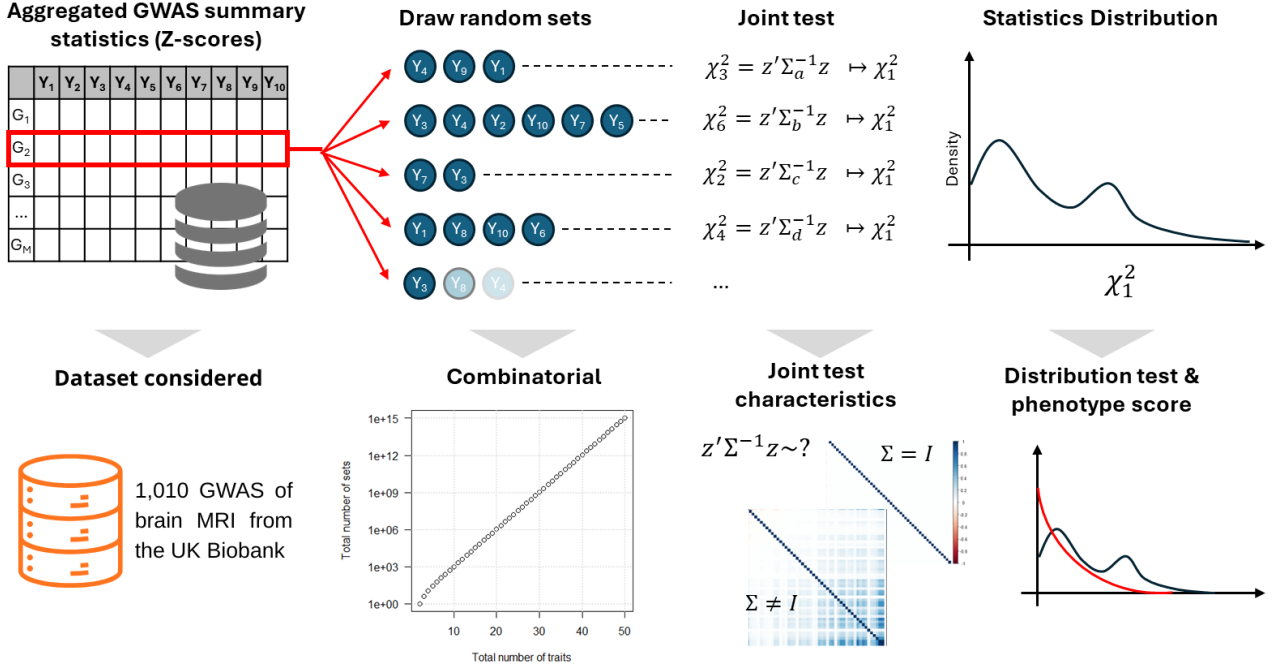


Figure 3: Pipeline illustration

The general structure of the sampling method follows as; rather than enumerating all possible subsets for K phenotypes, N_s random subsets of size k are drawn from the pool of K phenotypes. Sections 5, 6 and 7 follow the same **five-step pipeline** below, differing only in the dataset scale, the value of N_s and the range of k , and the covariance estimator used in Step 2. Sections 3 and 4 differ in the methods within certain steps. All these differences are documented within each section, while the shared computational core is defined once here:

Step 1. Input. The starting point at every stage is a z-score matrix $Z \in \mathbb{R}^{M \times K}$, the GWAS summary statistics described in Section 1.4: M rows, one per SNP, and K columns, one per phenotype, with entry $z_{j,k} = \hat{\beta}_{j,k}/\text{SE}(\hat{\beta}_{j,k})$. Some entries are missing; handling these is addressed from Section 5.1 onward, where affected SNPs are removed prior to sampling. This removal is by row, not by entry: a SNP with a missing value in even one phenotype column is dropped in full. Consequently, every Z_{sub} used from Section 5.1 onward and the full Z built in Section 7, is 0% sparse by construction: no NaN or null entries remain for Step 2 to draw phenotype subsets from. Each stage of this report operates on a working submatrix Z_{sub} of increasing SNP count and phenotype count, as detailed in that section's Dataset subsection. The full z-score matrix is implemented in Section 7.

Step 2. Sampling. For each of N_s subsets, a size k is drawn (uniformly, from a range specified per stage) and k phenotypes are selected without replacement from the K columns of Z_{sub} .

Step 3. Omnibus statistic. For a subset of k phenotypes, the joint test asks whether the SNP has an effect on any phenotype in that subset. Writing $\beta = (\beta_1, \dots, \beta_k)^\top$ for the effect of the SNP across the k selected phenotypes, the hypotheses are

$$H_0 : \beta = 0 \quad \text{vs} \quad H_1 : \beta_j \neq 0 \text{ for at least one } j,$$

compared to the univariate case, where $H_0 : \beta = 0$ is tested against $H_1 : \beta \neq 0$ for a single phenotype. The alternative here is therefore composite: rejecting H_0 establishes that the subset carries signal, but does not identify which of the k phenotypes is responsible.

Therefore, for a subset of k phenotypes with z-score vector $\mathbf{z} \in \mathbb{R}^k$, the omnibus test statistic is the Mahalanobis distance [3]

$$S = \mathbf{z}^\top \hat{\Sigma}_k^{-1} \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(0, \Sigma_k) \text{ under } H_0 \implies S \sim \chi_k^2,$$

where $\hat{\Sigma}_k$ is the $k \times k$ submatrix of the stage's covariance estimator restricted to the selected phenotypes (see Appendix 10.1 for why the LDSC intercept matrix, once introduced in Section 5, is the correct and unbiased choice of Σ_K).

Throughout this pipeline, three simplifying assumptions hold: since all 1,010 phenotypes derive from the same UKB cohort, the sample-overlap coefficient $N_s^{(i,k)} / \sqrt{N_i N_k}$ equals 1 for every pair of traits, so each entry of $\hat{\Sigma}_K$ reduces to the phenotypic correlation $\rho_{i,k}$ i.e. $\hat{\Sigma}_K$ is the LDSC intercept (correlation) matrix of the K selected phenotypes; $\hat{\Sigma}_k$ is treated as identical for every SNP; and each z-score is standard normal under the null hypothesis.

Rather than forming $\hat{\Sigma}_k^{-1}$ explicitly at $O(k^3)$ cost, $\hat{\Sigma}_k$ is Cholesky-factorised as $\hat{\Sigma}_k = LL^\top$ [4], and the triangular system $LY = \mathbf{z}$ is solved for $Y = L^{-1}\mathbf{z}$ at $O(k^2)$ cost. Since $Y \sim \mathcal{N}(0, I_k)$ under H_0 , $S = \mathbf{z}^\top \hat{\Sigma}_k^{-1} \mathbf{z} = \|Y\|^2 \sim \chi_k^2$. Because L depends only on the selected phenotype subset, not on the SNP, it is computed once per subset and reused across all SNPs: arranging every SNP as a column of Z_{sub} gives a single batched triangular solve (one BLAS call) that produces Y for every SNP simultaneously, reducing the per-subset cost from $O(k^3 \times N_{\text{SNP}})$ to one $O(k^3)$ factorization plus one $O(k^2 \times N_{\text{SNP}})$ batched solve.

Step 4. Rescaling to a common scale. Subsets of different sizes k produce χ_k^2 statistics on different scales, not directly comparable. Each statistic is rescaled to its 1-degree-of-freedom equivalent via the p-value-preserving transform:

$$S_1 = F_{\chi_1^2}^{-1}(F_{\chi_k^2}(S)),$$

which satisfies $P(\chi_1^2 > S_1) = P(\chi_k^2 > S) = p$ by construction. P-values are clipped to $[10^{-300}, 1]$ before the inverse transformation for numerical stability. (See Appendix 10.3 for more details)

Two equivalent implementations are used across this report, differing only in which SciPy API supplies the survival and inverse-survival functions:

- **Sections 3–5:** `chi2.isf(np.clip(chi2.sf(S, k), 1e-300, 1.0), 1)` (`scipy.stats`).
- **Section 6 onward:** `chdtri(1, chdtrc(k, S))` (`scipy.special`) — faster, bypassing the `scipy.stats.chi2` object-wrapper overhead.

Step 5. Output. The procedure yields two matrices, both indexed by subset along the first axis:

- **stat_matrix** $\in \mathbb{R}^{N_s \times N_{\text{SNP}}}$: the rescaled statistic S_1 from Step 4 above i.e., each subset's raw χ_k^2 association strength for a given SNP, converted onto the common χ_1^2 scale so that subsets of different sizes k remain comparable. Entry (n, j) is this value for SNP j under subset n : larger values indicate stronger evidence that the phenotypes in subset n , taken jointly, are associated with SNP j . Since $N_s \times N_{\text{SNP}}$ can reach the tens of billions of entries at full scale, this matrix is stored at float32 precision as a memory-mapped file rather than held in RAM (from Section 5 onward).
- **sel_matrix** $\in \{0, 1\}^{N_s \times K}$: the subset-membership matrix, with entry $(i, p) = 1$ if phenotype p was drawn in subset i and 0 otherwise, stored as `uint8`. Each row therefore contains exactly k_i ones, where k_i is the subset size drawn for subset i in Step 2.

Together, these two matrices are sufficient to reconstruct every subsequent analysis in this report without rerunning the sampling loop: **stat_matrix** supplies the per-SNP, per-subset association strength, and **sel_matrix** supplies the phenotype membership needed to attribute that strength back to individual phenotypes (Section 8).

From these two matrices, this procedure supports up to two further analyses:

- **Phenotype score.** For each SNP and each phenotype p , averaging the rescaled statistic S_1 over exactly the subsets that included p yields the **phenotype_score_matrix.dat** $\in \mathbb{R}^{N_{\text{SNP}} \times K}$. A phenotype scoring highly here consistently contributes strong multitrait signal across every subset it was drawn into, identifying traits that are reliably associated with a SNP regardless of which other phenotypes happened to accompany them in a given subset.
- **Density plot (Stability).** Estimating the distribution of the rescaled χ_1^2 statistics across all sampled subsets via kernel density estimation (KDE) across a number of replicates. The method visualizes how much multitrait signal random phenotype combinations typically produce for a SNP, providing a reference against which that SNP's observed (all-phenotype) statistic can be judged as unusually extreme.

Summary. This report follows a single method: random subset sampling for multitrait GWAS testing through five stages of increasing scale, from a five-phenotype proof of concept to the full 1,010-phenotype, 9.6-million-SNP dataset. The motivation throughout is the same: a SNP's association with a coordinated set of correlated brain phenotypes can be diluted away in a univariate test and is intractable to test exhaustively in a joint one, so random subset sampling offers a tractable middle ground between the two. The goal of what follows is fourfold: implement this existing pipeline correctly, validate it at each scale before trusting it at the next, use it by Section 8 to show concretely where subset-based testing finds signal that standard univariate and joint approaches miss, and going further still, construct that subset directly through cumulative joint testing of score-ranked phenotypes, calibrated against a matched-size simulated null (Section 8.6), to establish which SNPs carry genuine multitrait signal that selection bias alone cannot explain, a finding then cross-checked directly against the earlier subset-based comparison (Section 8.6.1). With that framing in place, Section 3 begins the first of these five stages.

3 Pilot Analysis: 5000 SNPs over 5 Phenotypes (UKB_5000x5.ipynb)

3.1 Dataset

The input z-score data here is specifically from schizophrenia GWAS, filtered to SNPs with $p < 0.001$, yielding 5,249 SNPs across all 1,010 brain MRI phenotypes.

This notebook therefore restricts the analysis to the first five phenotypes in the full z-score matrix. With $K = 5$ phenotypes, the total number of non-empty subsets is $\sum_{k=1}^5 \binom{5}{k} = 31$, which is small enough to enumerate exhaustively. This tractable case serves as a proof of concept: the full pipeline can be validated against a known combinatorial space before the problem is scaled up and a subsampling strategy becomes necessary.

3.2 Covariance Estimator: Naive Correlation

For each of the 31 subsets, the omnibus statistic (Section 2.2, Step 3) is computed using $\hat{\Sigma}_k^{naive}$, the $k \times k$ submatrix, restricted to that subset's selected traits, with k ranging from 1 to 5, of the full empirical 5×5 correlation matrix $\hat{\Sigma}_K^{naive}$. This full matrix is formed directly from the z-scores across all SNPs (the sample correlation of the columns $z_{\cdot,k}$), *not* the LDSC intercept matrix, which is not introduced until Section 5. The following explains why this substitution matters.

Approximate covariance matrix. The covariance matrix $\hat{\Sigma}_K$ must reflect the correlation structure of the z-scores *under the null hypothesis* i.e., the covariance arising solely from **sample overlap** and **phenotypic correlation**, with no genetic signal. The empirical phenotypic correlation computed directly from the z-score matrix conflates null and causal SNPs: shared genetic effects inflate the off-diagonal entries, causing $\hat{\Sigma}_K^{-1}$ to be under-estimated and the test statistic S to be deflated, yielding conservative and underpowered tests. The correct estimator is derived from the LDSC matrix, which regresses the product $z_i z_j$ on LD scores across SNPs. The intercept of this regression isolates the null covariance (phenotypic correlation due to sample overlap) from the genetic covariance absorbed by the slope. The LDSC intercept matrix therefore gives an unbiased estimate of Σ_K and is from Section 5 onwards. The LDSC intercept matrix is the statistically correct choice and is incorporated in future notebooks. (See Section 10.1 for more details)

So in this section, the following computational results are inflated yet it is still valuable to continue the analyses.

Adjusted implementation details. This pilot deviates from the canonical pipeline of Section 2.2, both harmless at this tiny scale.

- First, Step 2 (Sampling) is replaced by exhaustive enumeration, as above, since all 31 subsets are enumerable directly rather than drawn at random.
- Second, Step 3 (Omnibus statistic) is computed here by explicitly forming $(\hat{\Sigma}_K^{naive})^{-1}$ via a Cholesky-based solve of $\Sigma_K^{naive} X = I$, rather than the $O(k^2)$ -per-SNP triangular-solve trick ($LY = \mathbf{z}$, $S = \|Y\|^2$) that avoids forming the inverse altogether; that optimization is introduced in Section 5.5, and only matters once k and N_{SNP} grow large enough for the cost difference to be material.

Each subset's omnibus statistic ($S = \mathbf{z}^\top (\hat{\Sigma}_K^{naive})^{-1} \mathbf{z}$) is then rescaled to its χ_1^2 equivalent following

Step 4 of Section 2.2. This allows their association signals to be compared directly across subsets of different sizes.

3.3 Results and Visualization

The distribution of χ^2_1 statistics across all 5,249 SNPs was visualized for each subset using KDE, producing one density curve per phenotype subset. These distributions provide a first look at the range of association signals captured by different phenotype combinations for this SNP set. Sets containing highly correlated phenotypes tend to produce smaller χ^2_1 values after decorrelation, while sets that combine more independent traits capture broader signal.

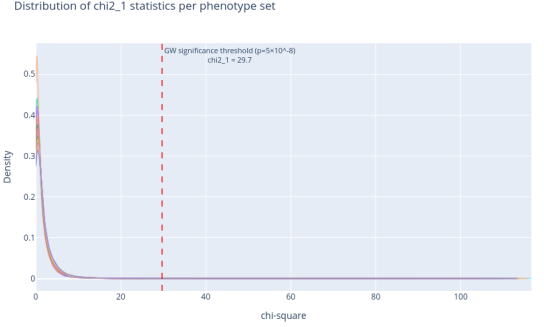


Figure 4: KDE of χ^2_1 statistics

Figure 4 shows the 31 curves are visually almost indistinguishable and heavily concentrated near zero, with essentially no density extending past the genome-wide significance threshold. This is not unexpected: these schizophrenia-filtered SNPs and the five somatosensory surface-area phenotypes were paired arbitrarily here to validate the pipeline’s mechanics, not because a genuine association was anticipated between them.

3.4 Limitations and Next Steps

Several aspects of this analysis are intentionally simplified:

- **Restricted phenotype pool.** Scaling to the full pool requires replacing exhaustive enumeration with the random subsampling strategy described in Section 2.2.
- **Per-SNP visualization.** The current plots aggregate over all SNPs. A more informative view would fix a single SNP of interest and overlay a vertical line at the full joint statistic, allowing direct assessment of whether that SNP exceeds the bulk of the sampling distribution.

4 Scaling Up: 100,000 SNPs over 10 Phenotypes (UKB_100kx10.ipynb)

4.1 Dataset

Here we randomly sampled 100,000 SNPs using stratified sampling.

Stratified sampling. To reduce runtime while demonstrating the pipeline at scale, 100,000 SNPs were drawn from the full 9.6M-SNP file via stratified sampling: the file is partitioned into 20 sequential batches of 500,000 rows, and $100,000/20 = 5,000$ rows are drawn uniformly at random from each (unique seed per batch), then concatenated. This bounds memory to one 500,000-row batch at a time. From the sampled SNPs, 10 phenotypes were selected uniformly at random from the pool of 1,010 (fixed seed), giving a working matrix of shape $100,000 \times 10$.

With $K = 10$ phenotypes, the total number of non-empty subsets is $\sum_{k=1}^{10} \binom{10}{k} = 1,023$, yielding $100,000 \times 1,023 = 102,300,000$ test statistics. Despite still being exhaustively enumerable, this 33-fold increase over the 5-phenotype pilot begins to illustrate the combinatorial pressure motivating a future

design decision: enforcing a minimum subset size (e.g. $k \geq 2$) to avoid trivial single-phenotype tests as the phenotype pool grows.

4.2 Covariance Estimator: JASS-inspired

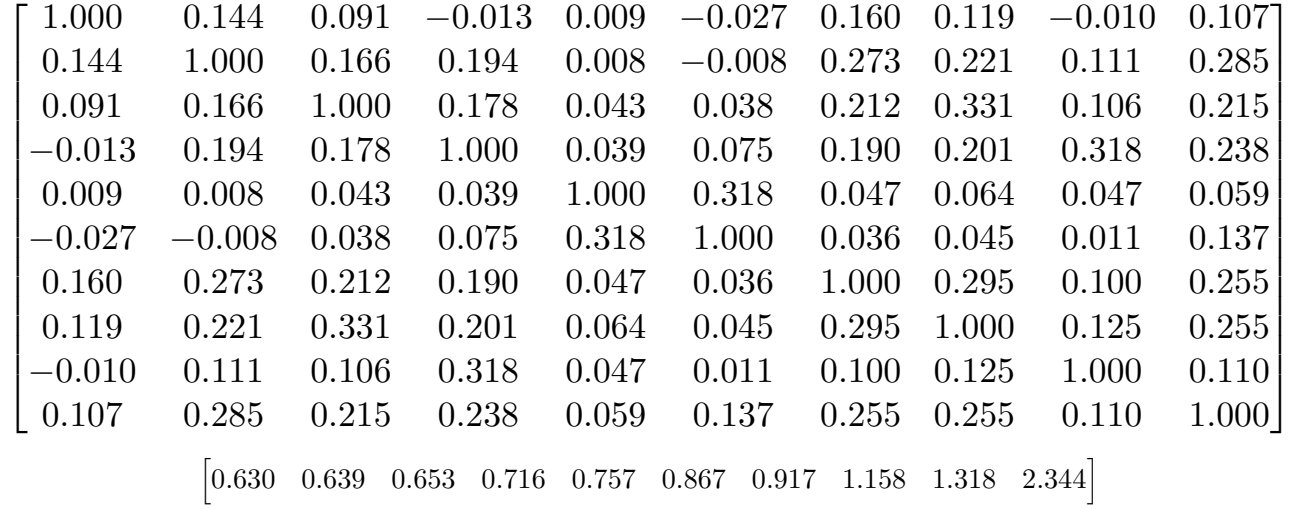


Figure 5: Correlation matrix (top) and its eigenvalues (bottom), 10 phenotypes. Minimum eigenvalue = 0.6299 > 0, confirming positive-definiteness.

Unlike Section 3’s naive empirical correlation, this notebook estimates Σ_K using an improved, null-SNP approach inspired by **JASS** (Joint Analysis of Summary Statistics) [5], a multitrait GWAS pipeline developed by the Statistical Genetics unit at Institut Pasteur. JASS restricts the correlation estimate to SNPs with weak marginal signal ($|z| < 1.96$ for every phenotype), on the reasoning that such SNPs are unlikely to carry true genetic effects, so their z-score correlation reflects only sample overlap and phenotypic correlation — the same quantity the LDSC intercept matrix targets directly. This notebook adopts the estimator $\hat{\Sigma}_K^{JASS}$ as an approximation to LDSC, which is introduced properly in Section 5. (See Section 10.2 for more details on the derivation)

This estimator plays the same conceptual role as the LDSC intercept matrix: it isolates the null covariance from the genetic signal, but it is not identical to LDSC. Nonetheless, this estimator serves as a computationally accessible proxy; $\hat{\Sigma}_K^{JASS}$ was verified to be positive definite by inspecting the eigenvalues. All eigenvalues are strictly positive.

As in Section 3.2, this notebook also enumerates subsets exhaustively (Step 2) and computes the omnibus statistic via an explicit $(\hat{\Sigma}_K^{JASS})^{-1}$ solve rather than the triangular-solve trick (Step 3); both deviations from Section 2.2 remain harmless at $K = 10$, for the same reasons given there.

For each of the 1,023 subsets of size $k \in \{1, \dots, 10\}$, the omnibus test ($S = \mathbf{z}^\top (\hat{\Sigma}_k^{JASS})^{-1} \mathbf{z} \sim \chi_k^2$) statistics were computed via vectorized `einsum` operations over the SNP axis. The rescaling procedure is identical to that described in Section 3.4. Applied to all $100,000 \times 1,023$ statistics, the transformation completed in approximately 11 seconds, producing a χ_1^2 matrix of shape $(100,000, 1,023)$. The disproportionate runtime relative to the approximate 2 seconds required for the joint statistic computation reflects the cost of evaluating the χ^2 survival and quantile functions element-wise across 10^8 values, and suggests that optimization methods will be necessary as the analysis scales further.

4.3 Results and Visualization

Best-set identification. For each SNP, the subset yielding the largest χ^2_1 statistic was identified via a column-wise argmax over the $(100,000 \times 1,023)$ statistic matrix; looking up that subset’s row in `sel_matrix` (Section 2.2) then gives a per-SNP summary of which phenotype combination drives the strongest association signal.

Replicate sampling visualization (Figure 6). The first implementation of the stability check described in Section 2.2 was carried out for a single fixed variant, `rs78794880`: subsets of size $k \in \{2, \dots, 10\}$ were pooled from the full 1,023, and $R = 30$ independent bootstrap replicates of 10,000 sets each were drawn with replacement from this pool, with a KDE computed per replicate.

The 30 curves overlap almost perfectly, indicating that 10,000 samples are enough to produce a stable empirical density for a single variant. This is a useful check before applying the sampling approach at genome-wide scale. The distribution is right-skewed, as expected under the null, with most subsets producing small statistics; a variant with genuine pleiotropic signal would instead show its red all-phenotype line shifted far into the right tail. Here the red line falls essentially at the mode, well within the bulk of what random subsets produce, suggesting `rs78794880`’s association (if any) is weak or phenotype-specific rather than broadly shared across the 10 brain MRI measures.

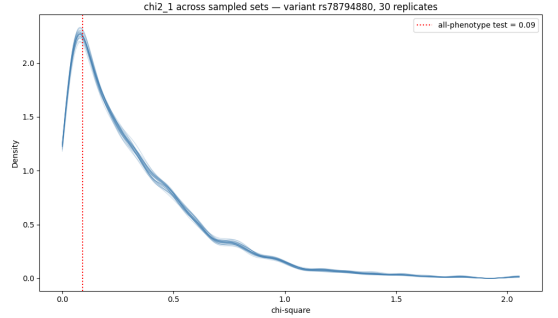


Figure 6: KDE of χ^2_1 estimates

Phenotype score (Figure 7) displays the per-phenotype average χ^2_1 statistic for variant `rs78794880` across 30 bootstrap replicates. Each box summarises the distribution of replicate-level averages for one phenotype; the tightly packed boxes indicate that 10,000 bootstrap samples per replicate were sufficient to produce stable per-phenotype score estimates.

`Volume_CSF_whole-brain` is the clear dominant contributor, with an average $\chi^2_1 \approx 0.51$ which is substantially higher than all other phenotypes and well-separated from the next ranked trait, `Volume_Hippocampus_right`. The remaining phenotypes cluster in a narrower band between 0.23 and 0.33, with `Area_rostralanterior_cingulate_left` producing the weakest average signal.

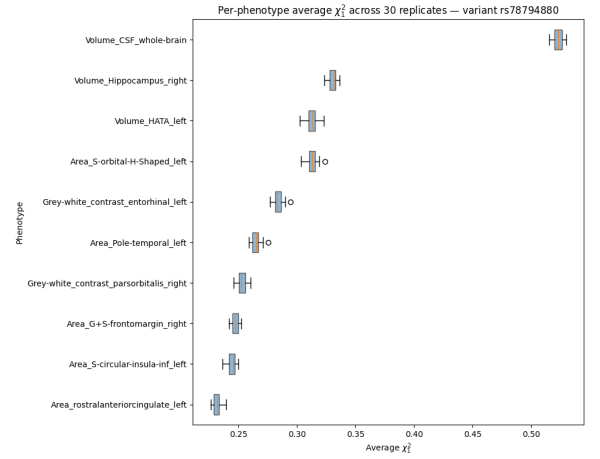


Figure 7: Phenotype score across 30 replicates

4.4 Limitations and Next Steps

Several aspects of this analysis remain simplified or warrant attention as the pipeline scales:

- **Approximate covariance estimator.** The low-signal null estimator is a practical proxy for the LDSC intercept matrix but not statistically equivalent; the LDSC intercept matrix had in fact already been pre-computed for these phenotypes, but was re-derived from scratch (covered

in Section 5).

- **Bootstrap vs. independent replication.** The 30 replicates in the visualisation are bootstrap resamples from a closed pool of 1,023 pre-computed values, not independent re-runs of the sampling procedure. True stability assessment requires independently drawn random subsets, as is the case once exhaustive enumeration is replaced by random sampling.
- **File format.** Intermediate outputs (joint statistics, χ^2_1 matrices) were saved as tab-separated CSV files, which becomes prohibitively slow and memory-intensive at larger scales; memory-mapped binary formats are adopted from Section 5 onward.
- **Matrix inversion at scale.** As K approaches 1,010, the cost of inverting $\hat{\Sigma}_K$ for every subset grows substantially; efficient large-scale inversion methods remain to be explored.
- **NaN values.** The z-score matrix may contain empty cells; rows with any missing value are dropped rather than zero-filled, since zero-filling would bias the joint test toward the null. This costs sample size but keeps estimates unbiased.

5 1,000,000 SNPs over 100 Phenotypes (UKB_1milx100.ipynb)

5.1 Dataset

Now, the input z-score matrix (`df_1mil.csv`) contains the first 1,000,000 SNPs across the first 100 brain MRI phenotypes from the UKB.

Nine SNPs contained at least one empty cell across the 100 phenotypes and were removed, leaving a working matrix $\mathbf{Z}$ of shape $999,991 \times 100$. This matrix was saved to disk as a binary `Z_matrix.npy` file and subsequently loaded via memory-mapping (`np.load(..., mmap_mode='r')`), which provides near-instant access to the array without reading the full file into RAM.

With $K = 100$ phenotypes, the total number of non-empty subsets is $\sum_{k=1}^{100} \binom{100}{k} = 2^{100} - 1 \approx 1.27 \times 10^{30}$, making exhaustive enumeration entirely infeasible.

5.2 Covariance Estimator: LDSC intercept matrix

This notebook introduces the statistically correct covariance estimator flagged as a limitation in Sections 3.2 and 4.2: LDSC intercept matrix Σ_{int} . As established in Appendix (Section 10.1), the LDSC estimator isolates the null covariance arising from sample overlap and phenotypic correlation by regressing the product $z_i z_j$ on LD scores across SNPs [6]; the intercept of this regression is uncontaminated by shared genetic effects. The values of the LDSC intercept matrix had been pre-computed for all 1,010 phenotypes and were stored as a $1,010 \times 1,010$ correlation matrix as `LDSC_full_matrix.csv`. It was then restricted to the 100×100 block corresponding to the first 100 selected phenotypes, giving $\hat{\Sigma}_{100}$ (Figure 8).

The matrix was verified to be positive definite by confirming that all eigenvalues of $\hat{\Sigma}_{100}$ are strictly positive, which guarantees that the Cholesky decomposition is valid. This matrix is visualized as a 100×100 heatmap (Figure 8).

We implement random sampling strategy described in Section 2.2: $N_s = 10,000$ subsets are drawn independently at random. For each subset $i \in \{1, \dots, N_s\}$, the subset size k_i is drawn uniformly from $\{5, \dots, 89\}$, and k_i phenotypes are then selected without replacement from the pool of 100. The lower bound of 5 is set to avoid trivially small subsets. The total number of test statistics computed is therefore $10,000 \times 999,991 \approx 10^{10}$. As to be described in Section 5.3, each subset requires only a single matrix factorization regardless of the number of SNPs.

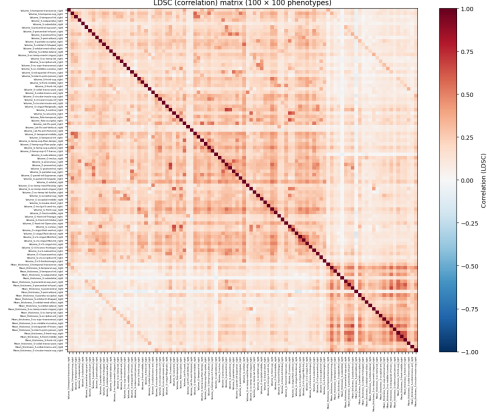


Figure 8: $\hat{\Sigma}_{100}$ heatmap

5.3 Sampling Loop

The per-subset computation: Cholesky factorization, batched triangular solve, and rescaling follows Section 2.2 exactly. What this section adds is the engineering needed to run that computation across all $N_s = 10,000$ subsets in parallel. Since each subset requires only a single $O(k^3)$ Cholesky factorization regardless of N_{SNP} (Section 2.2, Step 3), the natural axis to parallelize over is **subsets, not SNPs**. Hence growing the SNP count increases the cost of each subset’s batched triangular solve linearly, but does not change how many independent units of work there are to distribute across workers.

Parallelization and output. The 10,000 subsets were divided into 125 independent chunks of 80, distributed across 16 parallel workers. Each worker opened the shared memory-mapped file `stat_matrix.dat` in read-write mode and wrote its assigned rows directly to disk, avoiding the need to hold the full matrix in RAM. Upon completion, the results were assembled into the two output matrices described in Section 2.2, here of shape `stat_matrix` ($10,000 \times 999,991$) and `sel_matrix` ($10,000 \times 100$).

Both matrices were (casually) verified to be non-empty by inspecting a 20×20 block of `stat_matrix` and the first five rows of `sel_matrix`, confirming that all entries were finite and that each row of `sel_matrix` contained exactly k_i ones, consistent with the subset sizes drawn. The full job completed in 1 hour 41 minutes on 16 CPU cores with up to 128 GB RAM available.

5.4 Verification of the Cholesky Solver

Before running the full sampling loop, the Cholesky-based computation (Section 2.2, Step 3) was validated against a naive reference: for a single random subset (fixed seed) applied to the first 1,000 SNPs, $\hat{\Sigma}_k^{-1}$ was instead formed explicitly via `np.linalg.inv` and the statistic computed directly as $S_k(n) = \mathbf{z}_n^\top \hat{\Sigma}_k^{-1} \mathbf{z}_n$.

The maximum absolute difference between the two methods across all 1,000 SNPs was below 7×10^{-13} , and after rescaling both to χ_1^2 the outputs were identical. This confirms that the Cholesky solver is a numerically stable and correct substitute for explicit inversion, while avoiding both the $O(k^3)$ cost of forming $\hat{\Sigma}_k^{-1}$ and the instability explicit inversion introduces for near-singular matrices. The smallest eigenvalue of $\hat{\Sigma}_{100}$ is 0.225473, confirming positive-definiteness and that the Cholesky solver is valid here.

5.5 Stability of the Sampling Method

To assess whether $N_s = 10,000$ subsets is sufficient to produce stable estimates, a stability check was carried out for a single randomly chosen SNP (in this case: **rs10904727**). Rather than re-running the full 999,991-SNP job, the check operates on the z-score vector for a single SNP across all 100 phenotypes, $\mathbf{z}_{\text{snp}} \in \mathbb{R}^{100}$. Since only one statistic is computed per subset rather than 999,991, this is fast enough to run $N_{\text{rep}} = 10$ fully independent replicates of 10,000 subsets each. Each replicate uses a fresh random seed, so the replicates are genuinely independent re-runs of the sampling procedure rather than bootstrap resamples of a shared pool which directly addresses the limitation noted in Section 4.4.

For each replicate r , the mean and median χ_1^2 statistic across the 10,000 subsets were recorded. The results are summarized:

	Rep 0	Rep 1	Rep 2	Rep 3	Rep 4	Rep 5	Rep 6	Rep 7	Rep 8	Rep 9	Std across reps
Mean	0.1842	0.1832	0.1861	0.1816	0.1878	0.1873	0.1848	0.1851	0.1841	0.1833	0.0019
Median	0.0513	0.0532	0.0551	0.0538	0.0535	0.0547	0.0553	0.0557	0.0536	0.0529	0.0013

Figure 9: Mean and median χ_1^2 statistic across 10 independent replicates of 10,000 subsets for SNP **rs10904727** (index 907146), with the standard deviation of each summary statistic across replicates.

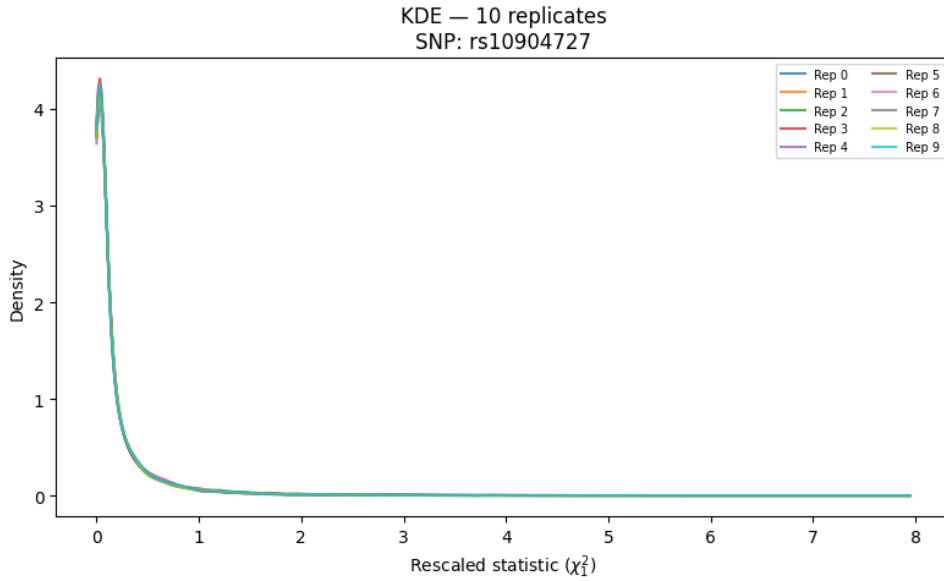


Figure 10: KDE of χ_1^2 statistics

Figure 10 shows that the replicates overlap closely, confirming that 10,000 subsets is sufficient to produce a stable empirical distribution for this SNP. Whether this holds uniformly across all SNPs may depend on the signal strength of the individual variant; SNPs with stronger multitrait signal may exhibit more variable distributions.

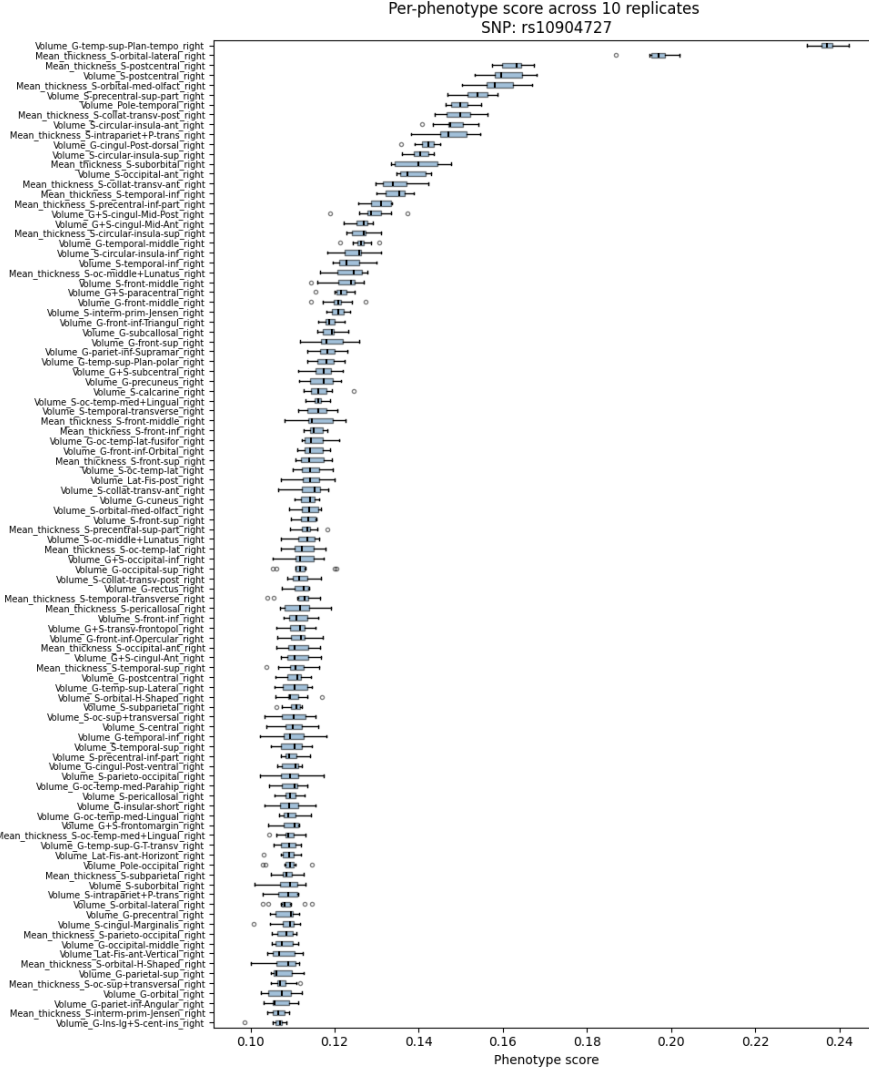


Figure 11: Phenotype score averaged across 10 replicates

Figure 11 shows the distribution of the average χ^2_1 statistics for each of the 100 phenotypes for when such phenotype is included in the randomly chosen subset. The top-ranked phenotype separates clearly from the rest, with a score around 0.23 against a smoothly declining band from ≈ 0.20 down to ≈ 0.10 for the remaining 99. This suggests that a single phenotype is driving most of this SNP's multitrait signal. This is only one SNP, however: a single dominant phenotype here neither confirms that this is a general property of the phenotype pool nor that it holds for other SNPs. Whether specific phenotypes (or combinations of them) consistently drive signal across many SNPs, or whether this pattern is idiosyncratic to rs10904727, would need to be checked systematically; a question taken up at full scale in Section 8.5

5.6 Limitations and Next Steps

- **Testing different N_s values.** The sampling loop was fixed at $N_s = 10,000$ subsets, which is a sparse sample of the $\sim 10^{30}$ combinatorial space so it is natural to increase. However, see Section 8.2 for more insight.
- **Restricted phenotype.** Only 100 of 1,010 phenotypes in the pipeline. Scaling to the full dataset requires further computational optimization.

6 Halfway there: 5,000,000 SNPs over 500 Phenotypes

6.1 Input Dataset

This analysis takes the first 5,000,000 rows (SNPs) sequentially from the full z-score table (`all_z_scores.txt`) and the first 500 phenotype columns, ordered to match the LDSC intercept matrix for consistent submatrix indexing, giving a matrix of shape $5,000,000 \times 500$.

The data were loaded via Polars lazy scanning with `float32` schema overrides on all phenotype columns, reducing the working dataset to ≈ 10 GB in RAM (see Glossary for the float32/float64 precision trade-off used throughout, including the survival-function underflow fix detailed in Section 6.3), and written to a Parquet file for efficient downstream access.

Rows containing any empty cells were removed prior to constructing the matrix: of the 5,000,000 SNPs, 59 contained at least one NaN z-score and were dropped, yielding a final matrix of shape $4,999,941 \times 500$, saved as `Z_matrix.npy`.

Main aim. The focus at this stage was to optimize the sampling loop’s runtime and reduce peak RAM usage and on-disk storage, rather than to perform downstream analysis of the outputs.

With $K = 500$ phenotypes, exhaustive enumeration is again intractable ($\approx 3.3 \times 10^{150}$ subsets), so this analysis adopts the random subsampling strategy of Section 5 with $N_s = 10,240$ subsets. Reproducibility was ensured by seeding a NumPy `SeedSequence`, spawning 320 independent child seeds, one per chunk; parallelization used `joblib` with 32 threads, each processing a chunk of 32 subsets ($320 \times 32 = 10,240$ total).

6.2 Sampling $\mapsto$ Omnibus Test $\mapsto$ Rescaling

The covariance matrix $\hat{\Sigma}_{500}$ is the 500×500 block matrix from the pre-computed LDSC intercept matrix introduced in Section 5, cast to `float32` to halve the per-worker memory footprint for the Cholesky factor and triangular solve relative to `float64`.

All 500 eigenvalues (via `numpy.linalg.eigvalsh`) are strictly positive (minimum ≈ 0.012 , maximum ≈ 79.2), confirming $\hat{\Sigma}_{500}$ is positive definite and the Cholesky factorization valid throughout the sampling loop.

Faster special functions. This notebook also replaces `chi2.sf/chi2.isf` (`scipy.stats`) with the faster `chdtrc/chdtri` (`scipy.special`) noted in Section 2.2, Step 4. The switch was verified equivalent on a randomly drawn subset of $k = 85$ phenotypes applied to the first 1,000 SNPs: maximum absolute difference 1.01×10^{-12} in the raw χ_k^2 statistic, and 3.77×10^{-13} after rescaling to χ_1^2 — well within floating-point precision.

For each of the $N_s = 10,240$ subsets of size $k \in \{5, \dots, 89\}$, the omnibus statistic (Section 2.2, Step 3) was computed simultaneously across all 4,999,941 SNPs via the Cholesky approach of Section 5.5, using $\hat{\Sigma}_K$, the $k \times k$ submatrix of $\hat{\Sigma}_{500}$ restricted to the selected phenotypes.

The z-score matrix was pre-transposed to a $500 \times 4,999,941$ C-contiguous array prior to the sampling loop, so that gathering the k selected phenotype rows via `Z_T[isSel, :]` produces a contiguous memory block rather than scattered reads, improving cache efficiency for the triangular solve; finite-value

checking was disabled (`check_finite=False`) in both the Cholesky factorization and the triangular solve, since NaN-containing rows were already removed during data preparation.

The output statistic matrix, shape $(10,240 \times 4,999,941)$ at `float32` precision (≈ 191 GB on disk), was written via a memory-mapped file (`stat_matrix.dat`), with each of the 32 worker threads writing exclusively to its own row indices, eliminating the need for synchronisation; the $(10,240 \times 500)$ `uint8` selection matrix was accumulated in RAM and saved separately as `sel_matrix.npy`. The full computation completed in approximately 4 hours 15 minutes on the `ggs` SLURM partition using 32 CPU cores with 150 GB of RAM.

The rescaling transformation (Section 2.2, Step 4) was applied inside the sampling loop immediately after each subset’s statistics were computed, operating on a vector of 4,999,941 values per subset rather than on the full $10,240 \times 4,999,941$ matrix at once, avoiding materializing the entire rescaled matrix in RAM. P-values were clipped to $[10^{-300}, 1]$ before the inverse transformation, and the rescaled statistics were cast to `float32` before being written to `stat_matrix.dat`, consistent with the `float32` memmap dtype. In total, approximately 51.2×10^9 statistics were rescaled across all $N_s = 10,240$ subsets.

6.3 Verification of the Parallel Sampling Loop (`verify_sampling.py`)

To validate the production sampling loop beyond the single-subset check in Section 5.4, a systematic verification compared the production method (**optimal**) against a **naïve** reference across $N_s = 100$ independently drawn subsets and 50,000 SNPs from the fullscale z-score matrix.

Naïve reference. For each subset, $S = \mathbf{z}^\top \hat{\Sigma}_K^{-1} \mathbf{z}$ was computed via explicit inversion (`np.linalg.inv`), and the rescaled χ_1^2 statistic via `chi2.sf/chi2.isf` (`scipy.stats`, always `float64` regardless of input precision). All linear algebra used `float32`, matching the production memory profile; this is intentionally the most straightforward implementation, using no Cholesky factorization or `scipy.special` functions.

Optimal method. The production implementation (`run_chunk` in `Sampling_method.py`) follows Section 2.2 exactly: Cholesky factorization, triangular solve, $S_k(j) = \|\mathbf{Y}_{:,j}\|^2$, and `chdtrc/chdtri` for rescaling, with `stat_all` cast to `float64` immediately before the survival-function call to avoid underflow, then back to `float32` on output.

Results. The selection matrix matched exactly across all 100 subsets, confirming reproducible random draws. For `stat_matrix.dat`, the maximum absolute difference in the rescaled χ_1^2 statistic was 7.229×10^{-4} (mean 1.261×10^{-6}), attributable solely to `float32` rounding between the two numerically distinct algorithms hence this difference is scientifically negligible.

Numerical issue identified. Passing `float32` `stat_all` to `chdtrc` without an explicit `float64` cast caused the survival function to underflow to exactly 0.0 for strongly associated SNPs, rendering the $[10^{-300}, 1]$ clip ineffective and propagating `inf` into the output. Casting `stat_all` to `float64` immediately before the `chdtrc` call fixed this with the unexpected side effect of also cutting runtime to 2 hours 39 minutes (same 32-core, 150 GB configuration).

6.4 Limitations and Next Steps

- **Memory scaling to the full dataset.** The sampling loop holds the full transposed z-score matrix $\mathbf{Z}_T$ ($500 \times 4,999,941$, ≈ 10 GB at `float32`) in RAM. Scaling to the full dataset (≈ 9.6 M SNPs

$\times 1,010$ phenotypes) would require ≈ 38 GB for this matrix alone, plus a $(k \times N_{\text{SNP}})$ submatrix per worker thread; at 32 threads with k up to 89, this peak footprint would exceed the available allocation and cause an out-of-memory kill. A SNP-batched inner loop is therefore introduced for the fullscale run (Section 7.2).

- **No downstream analysis.** `stat_matrix` and `sel_matrix` were not analysed here; per-SNP score aggregation and phenotype ranking are deferred to the fullscale z-score matrix.
- **Subset count.** $N_s = 10,240$ is a small fraction of the possible combinations; nonetheless $N_s = 10,000$ is retained in the fullscale configuration.

7 Fullscale: 9,611,261 SNPs by 1,010 Phenotypes

7.1 Input Dataset

Unlike the preceding notebooks, which worked from a subsample of SNPs, this stage processes the full z-score table (`all_z_scores.txt`, ≈ 82 GB, introduced in Section 2.2): all ≈ 9.6 million SNPs across all 1,010 brain MRI phenotypes.

The raw tab-separated file was streamed via Polars lazy scanning directly into a Parquet file, with phenotype columns reordered to match the LDSC intercept matrix (for consistent submatrix indexing) and cast to `float32`. Conversion was verified by comparing the first 10,000 rows cell-by-cell against the original CSV (`Check_matrices.py`): column order, SNP identifiers, and z-scores all matched exactly.

Rows containing any missing phenotype value were removed: of 9,611,405 SNPs, 144 (0.0015%) contained at least one NaN z-score and were dropped, leaving a final working matrix of shape $9,611,261 \times 1,010$, written to disk as `Z_matrix.npy` and loaded via memory-mapping for the sampling procedure in Section 7.2.

7.2 Sampling $\mapsto$ Omnibus Test $\mapsto$ Rescaling

This notebook uses the full $\hat{\Sigma}_{1010}$ LDSC intercept matrix (Figure 12), requiring only reordering to match the z-score matrix’s phenotype columns. It was cast to `float32` to match the sampling loop’s precision.

Positive-definiteness of the full $1,010 \times 1,010$ matrix $\hat{\Sigma}_{1010}$ was verified directly here. $\hat{\Sigma}_{1010}$ is exactly symmetric ($\max |\hat{\Sigma} - \hat{\Sigma}^\top| = 0$ across all entries, including the diagonal) and, via eigendecomposition (`numpy.linalg.eigvalsh`), strictly positive definite: all 1,010 eigenvalues are positive, with minimum 1.424×10^{-3} , maximum 164.03, and condition number 1.15×10^5 . This directly confirms the Cholesky factorization used throughout the sampling loop is valid for every subset drawn from the full phenotype pool, and gives an empirical basis for the ridge term introduced in Section 8.6: its 10^{-4} diagonal addition sits roughly an order of magnitude below the smallest true eigenvalue, rather than an assumed value, and

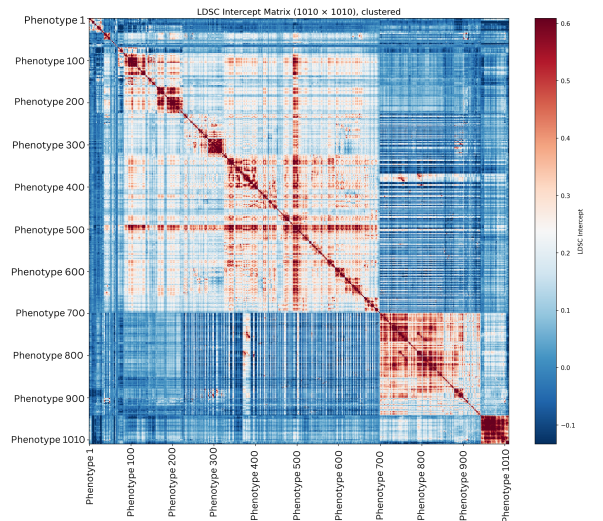


Figure 12: $\hat{\Sigma}_{1010}$ heatmap (clustered)

it is precisely this near-degeneracy, with condition number 1.15×10^5 , that motivates the eigenvalue-contamination check carried out in Section 8.6. As a secondary, now-redundant check, each subset's $k \times k$ submatrix is also Cholesky-factorized directly at the point of use (Section 7.4), which would fail immediately if not PD; the successful completion of all 10,000 factorizations is consistent with, though no longer required to establish, the direct eigenvalue confirmation above.

With all $K = 1,010$ phenotypes now in scope, exhaustive enumeration remains astronomically infeasible ($\approx 10^{304}$ subsets, Section 2.1). This notebook applies the same random subsampling strategy used since Section 5: $N_s = 10,000$ subsets were drawn, each of size k sampled uniformly from $\{5, \dots, 90\}$, with phenotypes drawn without replacement from the full pool of 1,010.

Reproducibility was ensured via a NumPy `SeedSequence` (fixed integer 7) spawning 100 independent child seeds, one per chunk. The 10,000 subsets were divided into 100 chunks of 100, distributed across 64 worker threads via `joblib` (`Parallel(n_jobs=64, prefer='threads')`). Since 100 chunks does not divide evenly across 64 workers, `joblib`'s dynamic queue means each thread pulls the next unprocessed chunk as soon as it finishes one: exactly 36 of the 64 workers end up processing a second chunk ($64 + 36 = 100$), the remaining 28 process only one, and which workers get a second chunk depends on runtime rather than a fixed assignment.

The per-subset computation follows Section 2.2 unchanged (Cholesky factorization, batched triangular solve, $S_k(j) = \|Y_{:,j}\|^2$). At full scale, however, holding the entire $(k \times M)$ arrays Z_{sub} , Y , and a Y^2 temporary in memory simultaneously, as in Sections 4 and 5, is no longer feasible: with $M \approx 9.6$ million SNPs and k up to 90, this would require tens of gigabytes per subset per worker thread, exceeding the allocation across 64 concurrent workers (the limitation flagged in Section 6.4). This notebook therefore introduces a **SNP-batched inner loop**, processing SNPs in contiguous blocks of `SNP_BATCH = 2,000,000` columns at a time, capping the per-thread memory footprint to $(k \times \text{SNP_BATCH})$ floats regardless of the total SNP count.

Since the Cholesky factor L depends only on the phenotype subset, not the SNP values, it is computed once per subset **outside** the batch loop and reused across all five batches needed to cover the 9.6 million SNPs which is exactly 10,000 factorizations and 50,000 batched triangular solves in total, not a per-SNP loop. Within each batch, the squared row norms are accumulated directly into a pre-allocated float32 buffer via `np.einsum('ij,ij->j', Y_batch, Y_batch, out=stat_all[b_start:b_end])`, avoiding the intermediate Y^2 array a naive `(Y**2).sum(axis=0)` would materialize; selected phenotype indices are also sorted in ascending order beforehand, improving cache locality when gathering the corresponding covariance-matrix rows.

Computing all $10,000 \times 9,611,261 \approx 9.6 \times 10^{10}$ test statistics this way, across the full parallel sampling loop, completed in 5 hours 17 minutes on 64 CPU cores, with peak memory usage of ≈ 100 GB (well under the 180 GB allocation).

The rescaling transformation (Section 2.2, Step 4) is applied inside the sampling loop immediately after each subset's statistics are computed, operating on the full $N \approx 9.6$ million-SNP vector `stat_all` at once rather than on the full $(10,000 \times 9,611,261)$ output matrix. Before the survival function is evaluated, `stat_all` is explicitly cast from float32 to float64: for strongly associated SNPs the tail probability $P(\chi_k^2 > S_k)$ can fall below 10^{-38} , which underflows to exactly 0.0 in float32 and propagates `inf` into the rescaled output (first identified in Section 6.3). P-values are then clipped to $[10^{-300}, 1]$

before the inverse transformation, and the resulting float64 statistic is cast back to float32 on assignment into the `stat_matrix.dat` memmap.

7.3 Verification of the Parallel Sampling Loop (`verify_sampling_fullscale.py`)

To validate the fullscale production sampling loop, a systematic verification compared the production method against the same naive reference used in Section 6.3, across $N_s = 100$ subsets, 50,000 SNPs, and all 1,010 phenotypes (5,000,000 pairwise comparisons). The SNP batch size was set to 5,000 (10 batches per subset) to exercise the batch loop at reduced scale before the full $N_{\text{SNP}} \approx 9.6$ million run.

Results. The selection matrix matched exactly across all 100 subsets. The maximum absolute difference in the rescaled χ^2_1 statistic was 7.52×10^{-4} (mean 1.24×10^{-6}), consistent with the float32 rounding observed in Section 6.3 and scientifically negligible.

Numerical issue identified. A single bug was uncovered and resolved: float32 underflow in the survival function, the same failure mode as Section 6.4 and fixed the same way (casting `stat_all` to float64 before `chdtrc`). Because $1e-300$ itself cannot be represented in float32, the intended $[10^{-300}, 1]$ clip was silently ineffective whenever `chdtrc` underflowed in float32, pinning affected SNPs to the clip ceiling (≈ 1373.87 on the rescaled χ^2_1 scale) rather than propagating `inf`.

This fix was applied to the production script `Sampling_method_fullscale.py` before the fullscale run.

7.4 Post-Run Data Integrity Check

Given the increased scale, a naive full scan of the outputs is not practical: `stat_matrix.dat` occupies approximately 384 GB on disk. Instead, `sel_matrix.npy` (9.7 MB) was loaded and checked in full, while `stat_matrix.dat` was checked via a random sample of 500 of the 10,000 subsets (rows) and 10,000 of the 9,611,261 SNPs (columns), drawn with a fixed seed, giving a $500 \times 10,000$ (20 MB) sample.

The full `sel_matrix` contained no NaNs and only the expected values $\{0, 1\}$. The sampled block of `stat_matrix` contained zero NaNs, zero infinities, and zero exact-zero entries, with sampled values ranging from 0.000000 to 840.011414. This gives high confidence that both output matrices are well-formed at full scale and free of the float32 underflow/overflow failure modes identified during verification.

7.5 Limitations and Next Steps

- **Fixed subset count and size.** $N_s = 10,000$ with subset size 5 to 90 remains a vanishingly small coverage of the $\sim 10^{304}$ combinatorial space established in Section 2.1. Section 8.2 shows that per-phenotype *average* scores converge well by this point, but extreme-value statistics used later (e.g. the max-based subset p-value in Section 8.3) do not carry the same convergence guarantee: a larger N_s could still reveal additional signal in the tail of the sampling distribution that 10,000 draws simply missed.
- **Purely uniform random sampling.** Subset size and membership are drawn uniformly at random at every scale in this report. An adaptive or informed sampling scheme could be explored.

8 Fullscale: Downstream Analysis

8.1 Phenotype Score Matrix Computation

The sampling procedure in Section 7 yields, for every SNP, up to $N_s = 10,000$ independent estimates of association strength, one per randomly drawn phenotype subset, each already rescaled onto the common χ_1^2 scale (`stat_matrix.dat`). To summarise this information per phenotype rather than per subset, this analysis computes a **phenotype score**: for a given SNP and a given phenotype, the average of the rescaled statistic S_1 over only those subsets that happened to include that phenotype.

Computation. Let `sel_matrix.npy` $\in \{0, 1\}^{N_s \times N_p}$ denote the subset-selection matrix ($N_p = 1,010$), where $\text{sel}[n, p] = 1$ if phenotype p was drawn in subset n , and let `stat_matrix.dat` $\in \mathbb{R}^{N_s \times N_{\text{snp}}}$ be the matrix of rescaled statistics, one row per subset and one column per SNP ($N_{\text{snp}} = 9,611,261$). Then, the phenotype **score_matrix.dat** $\in \mathbb{R}^{N_{\text{snp}} \times N_p}$ is:

$$\text{score}[j, p] = \frac{\sum_{n=1}^{N_s} \text{stat}[n, j] \cdot \text{sel}[n, p]}{\sum_{n=1}^{N_s} \text{sel}[n, p]},$$

i.e. the mean rescaled statistic at SNP j , averaged only over the subsets in which phenotype p participated. The denominator, $\text{counts}[p] = \sum_n \text{sel}[n, p]$, depends only on the sampling scheme and is computed once. The numerator, for a block of SNPs, is a single matrix product: `stat_matrix.T sel_matrix`, exploiting the fact that multiplying by the binary selection matrix and dividing by the per-phenotype counts reduces the sum to only the selected subsets.

Parallelisation and output. `Compute_score_matrix.py` splits the 9,611,261 SNPs into 4 contiguous, equally sized ranges, processed by 4 worker processes (`joblib`, `loky` backend), each pinned to 8 BLAS threads. Within a worker, its SNP range is further split into batches of `SNP_BATCH = 500,000` columns, bounding the memory footprint of the batch's slice of `stat_matrix.dat` to roughly $10,000 \times 500,000 \times 4 \text{ bytes} \approx 20 \text{ GB}$ per worker ($\approx 80 \text{ GB}$ peak across all 4 workers). Both matrices are read from and written to memory-mapped files, so no single array holding the full $N_{\text{snp}} \times N_p$ output is ever materialised in RAM; the output `score_matrix.dat` has shape (9,611,261, 1,010) in `float32`, occupying $\approx 38.8 \text{ GB}$ on disk.

Results. All four workers completed in 9.3 minutes, each processing its approximately ≈ 2.4 million-SNP range in five batches. This is markedly faster than the sampling stage itself (Section 7), reflecting that the score computation is a single pass of dense matrix multiplication over already-computed statistics, with no per-subset factorization or survival-function evaluation.

8.2 Robustness of the Score Estimates

The phenotype score computed in Section 8.1 is itself a sample average over only $N_s = 10,000$ random subsets. Since Section 7.5 already flags this as a fixed, comparatively small sample of the combinatorial space, this subsection checks empirically whether we can take less than 10,000 subsets for the phenotype score to have converged to a stable value and see if it is substantially noisy.

Methodology. For a single SNP, $N_{\text{rep}} = 10$ independent replicates of the production sampling procedure ($k \sim \text{Uniform}\{5, \dots, 90\}$, phenotypes drawn without replacement) were run sequentially, each replicate using an independently seeded generator (`np.random.SeedSequence(0)`, spawned once per replicate). Within a replicate, subsets are drawn one at a time and a running per-phenotype average score is maintained; this running average is checkpointed at $N_s \in \{1,000, 5,000, 7,500, 9,000, 10,000\}$ subsets. This yields, for every phenotype and every checkpoint, 10 independent estimates of the phenotype score at that sample size, whose spread directly measures how much the estimate would have varied had a different random draw of subsets been used.

SNP **rs80170726** was randomly selected for a closer, single-SNP inspection covering all 1,010 phenotypes as well as a restricted view of only the 10 phenotypes with the highest score at $N_s = 10,000$.

Results. Figure 13 shows, for **rs80170726**, boxplots of the 10 replicate score estimates at each checkpoint, for its top-10 scoring phenotypes. Across all of them, the spread of the boxes shrinks substantially from $N_s = 1,000$ to $N_s = 10,000$, and the medians stabilise well before the full 10,000 subsets are reached. The same pattern also holds across the full 1,010-phenotype view.

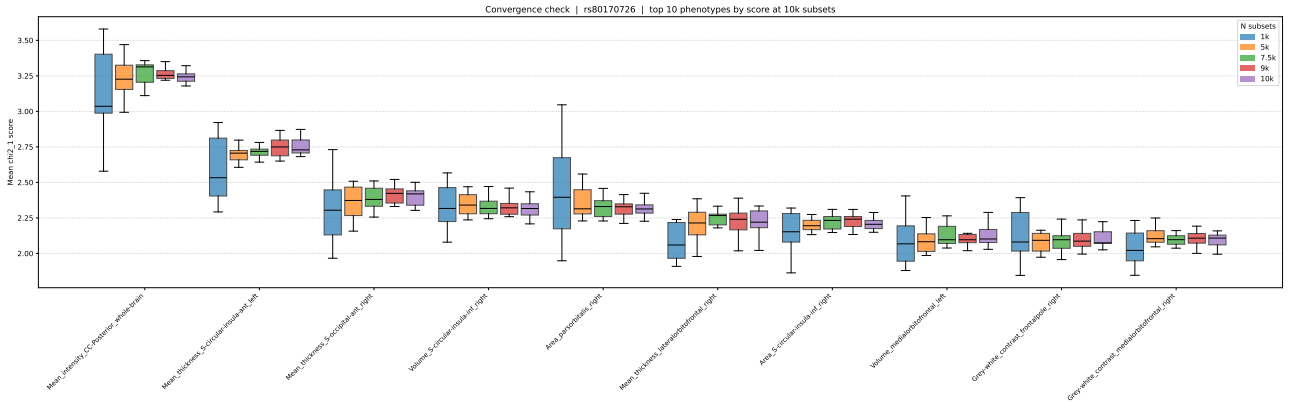


Figure 13: Convergence of the per-phenotype score estimate for SNP **rs80170726**, top 10 phenotypes by score at $N_s = 10,000$. Each box summarises 10 independent replicates of the sampling procedure, checkpointed at increasing numbers of subsets N_s .

Limitations. This check used only $N_{\text{rep}} = 10$ replicates. It also recomputes the sampling procedure independently rather than reusing `stat_matrix.dat` and `sel_matrix.npy` from the production run, so it verifies the convergence *behaviour* of the method rather than the specific values already stored on disk. It supports, but does not by itself prove, that $N_s = 10,000$ is an adequate sample size at every SNP.

8.3 Comparing Univariate, Joint, and Subset-Based P-values

With both the raw z-scores (`Z_matrix.npy`) and the fullscale sampling output (`stat_matrix.dat`, Section 7) available, this analysis compares three different per-SNP significance tests against one another: **standard single-phenotype univariate test**, a **joint test using all 1,010 phenotypes simultaneously**, and a **Bonferroni-corrected test based on the best of the 10,000 random subsets**.

Aim. To see whether the random subset sampling strategy is finding association signal that the other two, more standard, approaches miss.

The three tests. For each SNP j , three p-values are computed:

- **Minimum univariate p-value.** Each phenotype k is tested independently: $(z_{j,k})^2 \sim \chi_1^2$ under the null i.e, $\mu_{j,k} = 0$ (Appendix 10.2), giving $p_{j,k} = \text{chdtrc}(1, (z_{j,k})^2)$. The reported value is $\min_k p_{j,k}$, the strongest single-phenotype association for SNP j . This ignores correlations between phenotypes entirely.
- **Joint p-value.** The full 1,010-length z-score vector z_j is decorrelated using the Cholesky factor L of the LDSC intercept matrix (the same matrix and factorization approach as Section 7.2), $Y_j = L^{-1}z_j$, and $S_j = \|Y_j\|^2 \sim \chi_{1010}^2$ under the null.
- **Minimum subset p-value (Bonferroni-corrected).** For SNP j , the maximum of the 10,000 rescaled χ_1^2 -equivalent statistics already stored in `stat_matrix.dat` is taken, converted back to a p-value, and multiplied by 10,000 (clipped at 1) to correct for having tested 10,000 subsets per SNP [7]. This answers whether *any* sampled subset gives a significant result for SNP j , once the multiple-testing penalty for trying 10,000 of them is applied.

Computation. `Compute_pvalue_matrix.py` computes all three columns in a single combined pass over 4 worker processes (8 BLAS threads each), each handling a contiguous quarter of the 9,611,261 SNPs in batches of `SNP_BATCH = 200,000`. Both `Z_matrix.npy` and `stat_matrix.dat` are read within the same batch loop rather than in two separate scans, halving the I/O relative to computing the univariate/joint and subset-based columns independently. The run completed in around 124 minutes.

Results. At the conventional genome-wide significance threshold $p < 5 \times 10^{-8}$:

Criterion	SNPs
Significant in ≥ 1 of the 3 tests	1,944,582
Significant in ≥ 2 of the 3 tests	160,145
Significant in all 3 tests	64,718 (0.673% of 9,611,261)

Figure 14 is a **Q-Q plot**: for each SNP, the joint and minimum-subset $-\log_{10}$ p-values are plotted against the corresponding univariate $-\log_{10}$ p-value, with the $y = x$ line as the reference against which points are read as more or less significant. Points above the diagonal indicate a more significant result than the best single phenotype. We can see most points fall above the $y = x$ diagonal for the joint test, indicating that decorrelating and combining all 1,010 phenotypes typically yields a more significant result than the best single phenotype alone. The minimum-subset test falls closer on average to the diagonal, consistent with it being a noisier, sampling-based approximation of the same idea restricted to 10,000 random combinations rather than the single full combination. This gives the impression that there exist subset combinations that

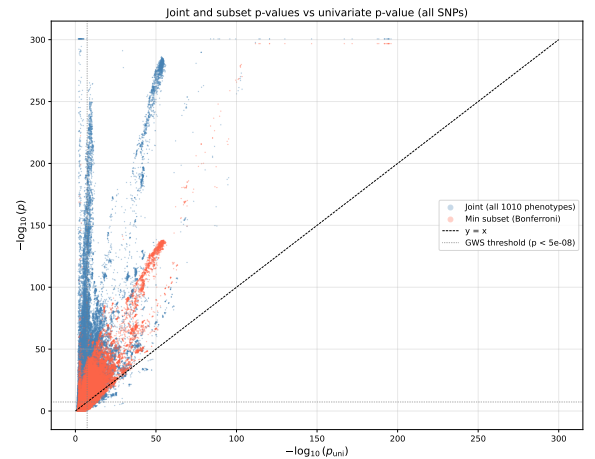


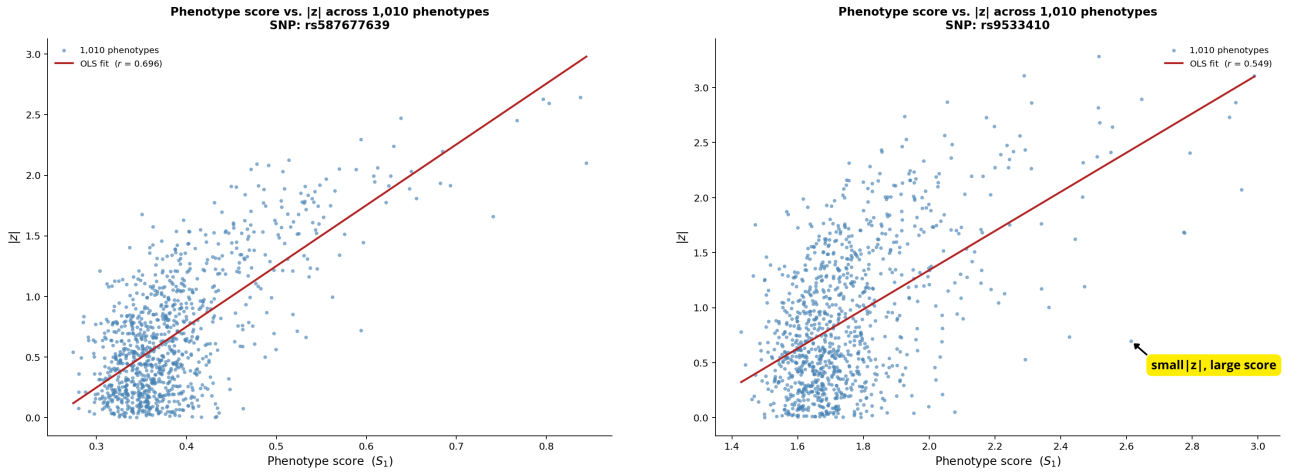
Figure 14: Q-Q-style comparison: $-\log_{10}$ p-values of joint and minimum-subset (Bonferroni-corrected) plotted per SNP against the minimum univariate

can push stronger signals that are not covered by our current 10,000 random subsets and so increasing N_s and subset size range is highly recommended.

Note. The minimum-subset test is Bonferroni-corrected for 10,000 subsets, which is conservative given that the subsets are highly overlapping in phenotype membership rather than independent tests. This means that the true effective number of tests is smaller, so this correction likely understates the significance of the minimum-subset p-value. This comparison also does not yet identify *which* phenotypes drive a SNP’s subset-based signal, or which SNPs are significant only under the subset-based test and not the other two — both addressed in the following subsection.

8.4 Phenotype Score vs. Univariate Signal

Aim. To assess whether the phenotype score captures association signal beyond what univariate testing detects, and in particular whether any phenotypes score highly in the subset test despite weak marginal z-scores.



(a) Expected pattern (SNP **rs587677639**, $r = 0.696$): phenotype score and $|z|$ are tightly correlated across the 1,010 phenotypes.

(b) Discordant pattern ($r = 0.549$): a subset of phenotypes (lower right) carry a large phenotype score but a small $|z|$, indicating associations invisible to univariate testing.

Figure 15: Phenotype score vs. $|z|$ across the 1,010 brain MRI phenotypes for two representative SNPs. Each point is one phenotype; the OLS trend line is shown in red.

For a given SNP, each of the 1,010 phenotypes can be placed in a scatter plot by its phenotype score at that SNP against its marginal $|z|$ at that SNP. Figure 15a shows a representative SNP (**rs587677639**) where the two quantities have a moderately high correlation ($r = 0.696$): phenotypes that score highly in the subset test also tend to have a strong univariate signal at that SNP.

Figure 15b shows a different SNP where this relationship breaks down ($r = 0.549$). A cluster of phenotypes in the lower right of the plot has a large phenotype score but a small $|z|$: their association with this SNP is not detectable in a single-trait test, yet the subset test consistently assigns them an elevated score. These phenotypes are found by subset testing precisely because their signal concentrates only when tested jointly with the right combination of other phenotypes. This is something neither a univariate test nor a joint test over all 1,010 phenotypes would reveal and so this clearly motivates testing phenotype subsets. This is the mechanism underlying the subset-specific discoveries characterized in: Section 8.5.

8.5 Characterizing Subset-Specific Discoveries

Section 8.3 compared the three tests in aggregate. This subsection zooms into the SNPs for which the subset-based test finds a significant result that the joint (all-1,010-phenotype) test does not. This is the clearest evidence that testing random phenotype combinations can surface signal invisible to a single test run on the full phenotype set and asks **which phenotypes are responsible for that signal**.

Overlap across the three tests. Table 1 breaks the full set of 9,611,261 SNPs down by every combination of significance ($p < 5 \times 10^{-8}$) across the three tests.

Category	SNPs (% of total)
None significant	7,666,679 (79.768%)
Univariate only	4,391 (0.046%)
Joint only	1,779,787 (18.518%)
Subsets only	259 (0.003%)
Univariate + Joint	12,805 (0.133%)
Univariate + Subsets	407 (0.004%)
Joint + Subsets	82,215 (0.855%)
All three	64,718 (0.673%)

Table 1: Contingency table of significance across the three tests, out of 9,611,261 SNPs.

The joint test alone accounts for the largest block of unique detections (18.518%), consistent with it being the most powerful of the three when a SNP affects many phenotypes concordantly. The subset-based test’s unique contribution is near zero: 259 SNPs reach significance only through the subset test, and a further 407 are also univariate-significant but still missed by the joint test.

Identifying subset-specific SNPs. Of the 147,599 SNPs significant under the min-subset-Bonferroni test, 666 (0.45%) are **subset-specific** (259 subsets-only + 407 univariate subsets): significant in the subset test but not in the joint test, irrespective of univariate status. These are SNPs for which some particular combination of phenotypes carries a signal that is diluted away once all 1,010 phenotypes are tested jointly.

Identifying phenotypes that drive the subset-specific signal. Two complementary, purely descriptive approaches were used to attribute each subset-specific SNP’s signal back to individual phenotypes:

- **By average score** (Table 2 - top). Using the phenotype score matrix from Section 8.1, the 20 highest-scoring phenotypes were identified for each of the 666 SNPs, and phenotype frequency was counted across this set of top-20 lists. A phenotype ranking highly here consistently receives a high average rescaled statistic across all subsets that included it, for these SNPs.
- **By winning subset** (Table 2 - bottom). For each of the 666 SNPs, the single best-performing subset out of the 10,000 (the one with the maximum χ^2_1 -equivalent statistic in `stat_matrix.dat`) was identified, and its phenotype membership was read directly from `sel_matrix.npy`. Phenotype frequency was then counted across these 666 winning subsets.

The two rankings (Table 2) do not agree on which phenotype is most implicated, since they measure different things: the average-score ranking reflects a phenotype’s consistent contribution across *every*

Phenotype (by average score)	% of 666 SNPs
Volume_Midbrain_whole-brain	15.32%
Volume_Putamen_right	13.66%
Volume_Putamen_left	13.06%
Mean_intensity_Cerebellum-Cortex_left	13.06%
Mean_intensity_Cerebellum-Cortex_right	12.16%
Phenotype (by winning subset)	% of 666 SNPs
Volume_SubCortGray_whole-brain	12.91%
Area_BA1_right	12.76%
Volume-ratio_BrainSegVol-to-eTIV_whole-brain	12.01%
Volume_L-Sg_right	11.56%
Volume_Cerebellum-Cortex_right	11.56%

Table 2: Top-ranked driver phenotypes for the 666 subset-specific SNPs, by average score (top) and by literal membership in each SNP’s winning subset (bottom).

subset it appears in, while the winning-subset ranking reflects only the *one* combination that happened to be most significant for each SNP.

Limitations. The 666-SNP definition used here does not exclude univariate-significant SNPs, so it does not isolate signal that is invisible to *every* standard test, only to the joint test specifically. The "top-20 by score" cutoff and the choice to look only at each SNP’s single best subset are both arbitrary analysis choices rather than statistically motivated thresholds; neither approach constitutes a formal test of which phenotype’s presence in a subset causes that subset’s significance, and the mean winning-subset size of 43 out of a possible 90 phenotypes means many other phenotypes are typically also present alongside the top-ranked ones. These analyses confirm the difficulty of finding specific phenotypes that are the primary drivers to push signal when implementing random subset sampling.

8.6 Cumulative Joint Testing of Score-Ranked Phenotypes

Aim. Section 8.5 attributed each subset-specific SNP’s signal back to individual phenotypes using two purely descriptive rankings: average score and winning-subset membership, which did not agree with one another and did not amount to a formal test of which phenotypes were responsible. This subsection introduces a third, constructive approach: rather than describing which phenotypes tend to score highly, the phenotypes are ranked by their score and then tested jointly in that order, building the subset directly instead of inferring it after the fact.

Methodology. For a given SNP, the 1,010 phenotypes are sorted in descending order of their phenotype score at that SNP (Section 8.1), giving an ordering $p_{(1)}, \dots, p_{(1010)}$. The omnibus statistic of Section 2.2 (Step 3) is then computed on each nested prefix of this ordering in turn: on $\{p_{(1)}\}$, then $\{p_{(1)}, p_{(2)}\}$, then $\{p_{(1)}, p_{(2)}, p_{(3)}\}$, and so on. For a prefix of length k the statistic $S_k \sim \chi_k^2$ under the null, converted to $-\log_{10}(p)$ as usual; the subset size at which this curve peaks is denoted peak k , and the corresponding prefix is taken as the subset of top-scoring phenotypes giving the strongest joint association for that SNP.

Computing each of the 1,010 nested statistics independently would require a fresh $O(k^3)$ Cholesky factorization at every prefix length $k = 1, \dots, 1010$, which is wasteful: the covariance submatrix Σ_{k-1}

for a prefix of size $k - 1$ is exactly the top-left $(k - 1) \times (k - 1)$ block of Σ_k , so almost all of the work done at step $k - 1$ could in principle be reused at step k rather than discarded. The following shows how to exploit this directly at the level of the Cholesky factor itself, so that the entire curve of 1,010 statistics for a SNP is built in a single incremental pass rather than 1,010 independent $O(k^3)$ factorizations.

Proposition (rank-one Cholesky update). Let $\Sigma_k \in \mathbb{R}^{k \times k}$ be symmetric positive definite, partitioned as

$$\Sigma_k = \begin{pmatrix} \Sigma_{k-1} & \Sigma_{1:k-1,k} \\ \Sigma_{1:k-1,k}^\top & \Sigma_{k,k} \end{pmatrix},$$

where $\Sigma_{k-1} \in \mathbb{R}^{(k-1) \times (k-1)}$ is the leading principal submatrix, $\Sigma_{1:k-1,k} \in \mathbb{R}^{k-1}$ the covariances between the newly added phenotype k and the $k - 1$ phenotypes already in the prefix, and $\Sigma_{k,k}$ the variance of phenotype k . Let L_{k-1} denote the unique lower-triangular, positive-diagonal Cholesky factor of Σ_{k-1} computed at the previous step. Then the Cholesky factor L_k of Σ_k has the block form

$$L_k = \begin{pmatrix} L_{k-1} & 0 \\ \boldsymbol{\ell}^\top & L_{k,k} \end{pmatrix},$$

i.e. its leading $(k - 1) \times (k - 1)$ block is exactly L_{k-1} , unchanged from the previous step, and only the new row $[\boldsymbol{\ell}^\top, L_{k,k}]$ needs to be computed.

Proof. Since Σ_k is symmetric positive definite, it admits a unique lower-triangular matrix L_k with strictly positive diagonal entries such that $\Sigma_k = L_k L_k^\top$ [4]. Partition L_k conformally with Σ_k :

$$L_k = \begin{pmatrix} A & B \\ \mathbf{c}^\top & d \end{pmatrix}, \quad A \in \mathbb{R}^{(k-1) \times (k-1)}, B \in \mathbb{R}^{(k-1) \times 1}, \mathbf{c} \in \mathbb{R}^{k-1}, d \in \mathbb{R}.$$

Because L_k is lower triangular, every entry strictly above the diagonal vanishes; in particular every entry of B lies in column k of a row $i \leq k - 1 < k$, hence strictly above the diagonal, so $B = 0$ identically. Writing $\boldsymbol{\ell} := \mathbf{c}$ and $L_{k,k} := d$, L_k therefore has the claimed block-triangular form. Expanding $L_k L_k^\top$ block by block,

$$L_k L_k^\top = \begin{pmatrix} A & 0 \\ \boldsymbol{\ell}^\top & L_{k,k} \end{pmatrix} \begin{pmatrix} A^\top & \boldsymbol{\ell} \\ 0^\top & L_{k,k} \end{pmatrix} = \begin{pmatrix} AA^\top & A\boldsymbol{\ell} \\ \boldsymbol{\ell}^\top A^\top & \boldsymbol{\ell}^\top \boldsymbol{\ell} + L_{k,k}^2 \end{pmatrix},$$

and equating this to Σ_k 's own block partition gives three equations:

$$AA^\top = \Sigma_{k-1}, \tag{i}$$

$$A\boldsymbol{\ell} = \Sigma_{1:k-1,k}, \tag{ii}$$

$$\boldsymbol{\ell}^\top \boldsymbol{\ell} + L_{k,k}^2 = \Sigma_{k,k}. \tag{iii}$$

Step 1 (identifying A with L_{k-1}). Equation (i) states only that A is *some* matrix satisfying $AA^\top = \Sigma_{k-1}$. But A is lower triangular (as the leading block of the lower-triangular L_k) and, since L_k has strictly positive diagonal by construction, A 's diagonal which are the first $k - 1$ diagonal entries of L_k , is strictly positive too. So A is *a* lower-triangular, positive-diagonal Cholesky factor of Σ_{k-1} . Since this factor is unique, and L_{k-1} is by definition the lower-triangular, positive-diagonal factor already satisfying $L_{k-1} L_{k-1}^\top = \Sigma_{k-1}$, uniqueness forces $A = L_{k-1}$ exactly: not merely a valid factor of Σ_{k-1} ,

but the identical matrix already computed at step $k - 1$, requiring no recomputation.

Step 2 (solving for ℓ). Substituting $A = L_{k-1}$ into (ii) gives $L_{k-1}\ell = \Sigma_{1:k-1,k}$. As L_{k-1} is lower triangular and already known, this is solved by forward substitution at $O(k^2)$ cost, versus the $O(k^3)$ a fresh factorization of Σ_k would require.

Step 3 (solving for $L_{k,k}$). Rearranging (iii) gives $L_{k,k} = \sqrt{\Sigma_{k,k} - \ell^\top \ell}$. This is well-defined precisely because Σ_k is positive definite: since $\ell = L_{k-1}^{-1} \Sigma_{1:k-1,k}$, the quantity $\ell^\top \ell = \Sigma_{1:k-1,k}^\top \Sigma_{k-1}^{-1} \Sigma_{1:k-1,k}$, so $\Sigma_{k,k} - \ell^\top \ell$ is exactly the Schur complement of Σ_{k-1} in Σ_k , and the Schur complement of a principal submatrix of a positive definite matrix is itself always positive. $\square$

Intuitively, $\Sigma_{k,k}$ is the total variance of phenotype k and $\ell^\top \ell$ is the portion of it already explained by its correlation with the $k - 1$ phenotypes already in the prefix, so $L_{k,k}^2$ is the variance in phenotype k left unexplained by them. Positive-definiteness of every Σ_k encountered along the score-ranked ordering; required for both the uniqueness argument above and the Schur-complement positivity in Step 3, is guaranteed by the ridge-regularized $\hat{\Sigma} + 10^{-4}\mathbf{I}$ used throughout this section, together with the eigenvalue bound established in Section 7 and the Cauchy-interlacing argument given below.

Updating the statistic without recomputing it. Recall from Section 2.2 (Step 3) that the omnibus statistic is obtained by solving $LY = z$ for $Y = L^{-1}z$ and setting $S_k = \|Y\|^2 = \sum_{i=1}^k Y_i^2$. Let $\mathbf{z}^{(k)} := (z_{(1)}, \dots, z_{(k)})^\top \in \mathbb{R}^k$ denote the z-score vector of the top- k score-ranked phenotypes, so that by construction of the nested ordering, $\mathbf{z}^{(k)}$ and $\mathbf{z}^{(k-1)}$ agree in their first $k - 1$ entries and $\mathbf{z}^{(k)}$'s final entry is $z_{(k)}$. Define $Y_k := L_k^{-1} \mathbf{z}^{(k)}$, the solution vector at prefix length k .

Corollary 1 (incremental update of Y). *Write $Y_k = (w_{1:k-1}^\top, \alpha_k)^\top$ with $w_{1:k-1} \in \mathbb{R}^{k-1}$. Then $w_{1:k-1} = Y_{k-1}$ exactly, and*

$$\alpha_k = \frac{z_{(k)} - \ell^\top Y_{k-1}}{L_{k,k}}.$$

Proof. Using $L_k Y_k = \mathbf{z}^{(k)}$ and the block form of L_k established in the Proposition above,

$$L_k Y_k = \begin{pmatrix} L_{k-1} & 0 \\ \ell^\top & L_{k,k} \end{pmatrix} \begin{pmatrix} w_{1:k-1} \\ \alpha_k \end{pmatrix} = \begin{pmatrix} L_{k-1} w_{1:k-1} \\ \ell^\top w_{1:k-1} + L_{k,k} \alpha_k \end{pmatrix} = \begin{pmatrix} \mathbf{z}^{(k-1)} \\ z_{(k)} \end{pmatrix} = \mathbf{z}^{(k)}.$$

Equating the leading $k - 1$ rows gives $L_{k-1} w_{1:k-1} = \mathbf{z}^{(k-1)}$ – exactly the linear system defining $Y_{k-1} = L_{k-1}^{-1} \mathbf{z}^{(k-1)}$. Since L_{k-1} is lower triangular with strictly positive diagonal entries, it is invertible, so this system has a unique solution; hence $w_{1:k-1} = Y_{k-1}$ is forced by uniqueness, not merely equal in value by relabeling. Equating the final row gives $\ell^\top w_{1:k-1} + L_{k,k} \alpha_k = z_{(k)}$, and substituting $w_{1:k-1} = Y_{k-1}$ and solving for α_k gives the claimed formula. $\square$

This is the same uniqueness argument used for $A = L_{k-1}$ in Step 1 of the **Proposition**, applied here to the solve rather than the factorization: just as the Cholesky factor's leading block cannot be some other valid factor of Σ_{k-1} , the solution vector's leading entries cannot be some other vector satisfying a similar-looking equation. Both are pinned down exactly by uniqueness of a nonsingular triangular system. The numerator of α_k is the residual of phenotype k 's z-score after subtracting the part predicted by the phenotypes already in the prefix ($\ell^\top Y_{k-1}$), and dividing by $L_{k,k}$ standardizes it to unit variance. This is why $\alpha_k \sim \mathcal{N}(0, 1)$ independently of $\alpha_1, \dots, \alpha_{k-1}$ under the null, established rigorously in the next paragraph. Moreover, since $S_k = \sum_{i=1}^k Y_i^2$ is simply a running sum and α_k is

exactly its new k -th term, the statistic itself updates as

$$S_k = S_{k-1} + \alpha_k^2,$$

with no need to recompute the terms already summed into S_{k-1} .

Update preserves the correct null distribution. As established in Section 2.2, under the null hypothesis $Y = L^{-1}z \sim \mathcal{N}(0, I_k)$: every entry of Y is an independent standard normal random variable and $\alpha_k \sim \mathcal{N}(0, 1)$ independent of $\alpha_1, \dots, \alpha_{k-1}$. A sum of k independent squared standard normals is by definition a chi-squared random variable with k degrees of freedom, so $S_k = \sum_{i=1}^k \alpha_i^2 \sim \chi_k^2$ under the null – exactly the distribution required for the cumulative test to be interpreted correctly at every prefix length, and consistent with the non-incremental derivation of $S_k \sim \chi_k^2$ given in Section 2.2. This guarantee holds for any fixed, data-independent ordering of phenotypes; the ordering actually used here is the SNP’s own score rank, which is data-dependent, so this result alone does not yet establish that the score-ranked p -values are well-calibrated. This gap is addressed directly in the **Null calibration** paragraph below.

Net effect. Walking through the score-ranked ordering once, updating L by one new row and S by one new squared term at each step, produces the entire cumulative curve $S_1, S_2, \dots, S_{1010}$ for a SNP at a cost of $O(k^2)$ per step rather than a fresh $O(k^3)$ factorization at every one of the 1,010 prefix lengths. This is the saving that makes the cumulative test tractable across all 9,611,261 SNPs.

A ridge of 10^{-4} was added to the diagonal of $\hat{\Sigma}$ before factorization since the smallest eigenvalue of the full $1,010 \times 1,010$ matrix is 1.424×10^{-3} (Section 7). This is an order of magnitude below the quantity it protects against and does not materially distort the covariance structure. This comparison is in fact the conservative, worst-case one: by **Cauchy’s interlacing theorem**, the smallest eigenvalue of any $k \times k$ principal submatrix of a symmetric matrix is always at least as large as the smallest eigenvalue of the full matrix, so no score-ranked prefix of any size k can have a smaller minimum eigenvalue than 1.424×10^{-3} . Hence the ridge is benchmarked against the hardest case the incremental Cholesky pipeline could ever encounter, not merely a representative one. Tail probabilities were evaluated via a wrapper, `_safe_log10p`, that calls `scipy.stats.chi2.logsf` and falls back to a normal approximation whenever `logsf` underflows to $-\infty$ which it does for sufficiently extreme statistics, since `scipy.stats.chi2.logsf` has no large-argument approximation of its own and simply returns $-\infty$ past a fixed threshold. This fallback was triggered for 129 of the 9,611,261 real SNPs and none of the simulated null SNPs. Those 129 peaks were recomputed exactly using the regularized upper incomplete gamma function at 80 digits of precision (`fix_fallback_peaks.py`). The largest correction, for `rs10699448`, moved an apparent peak of 32,614.47 down to an exact 330.45.

Early stopping. Evaluating every curve out to $k = 1,010$ is dominated by prefixes far beyond where the peak actually occurs, so the loop exits once the running statistic falls more than 30% below the highest value reached so far, subject to a floor of five phenotypes; the peak is then recorded as that running maximum. Across the full dataset this cuts the work to roughly 3% of the full curve. One consequence worth flagging explicitly: the rule returns the *first* local maximum whenever a curve declines and only later recovers, rather than the true global maximum which was the same rule and tolerance are applied identically to the null run, so both real and null curves are subject to the same possible under-detection, keeping the comparison fair even if not perfectly exhaustive.

Null calibration. The ranking used to order phenotypes is derived from the very z-scores that then enter the cumulative joint test, so the phenotypes are selected on the data and the resulting p-values cannot be taken at face value — the top-ranked prefix is, almost by definition, built from whatever happens to be most extreme in that SNP’s z-vector, and the joint statistic then partly rediscovers what the ranking already knew. To measure the size of this effect, a null dataset of the same dimensions as the real one was simulated: 9,611,261 z-vectors drawn as $z \sim \mathcal{N}(0, \hat{\Sigma}_{1010})$, carrying the real phenotype correlation structure but no genetic signal by construction. These were passed through the identical pipeline; the same 10,000 subsets from Section 7.2, the same omnibus test and rescaling, the same phenotype score computation of Section 8.1, and the same early-stopping rule above so that any peak this null run produces is attributable entirely to the selection procedure and nothing else. Each real SNP’s peak is then converted to an empirical p-value against this null distribution,

$$p_{\text{emp}} = \frac{1 + \#\{n : \text{peak}_n^{\text{null}} \geq \text{peak}^{\text{obs}}\}}{1 + N_{\text{null}}},$$

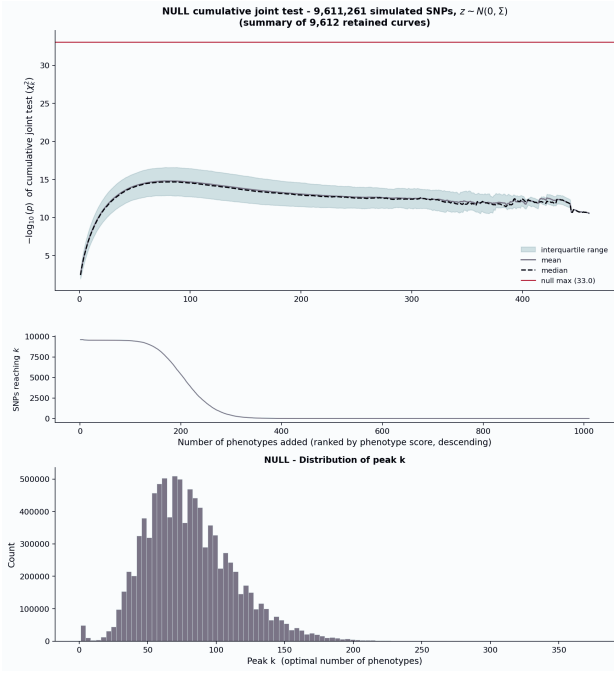
where the +1 offsets prevent an empirical p-value of exactly zero. With $N_{\text{null}} = 9,611,261$ the smallest attainable value is 1.04×10^{-7} , so SNPs exceeding the largest null peak are reported as a count rather than individually ranked; matching the null to the real dataset’s full size, rather than simulating a smaller reference sample, is precisely what brings this floor low enough to be useful at genome-wide scale — the expected number of real SNPs exceeding the null maximum by chance alone is $N_{\text{SNP}}/(N_{\text{null}} + 1) \approx 1$.

Results. Figure 16 shows the null run (a) and the real run (b) side by side, each summarised across three stacked panels: the top panel gives $-\log_{10}(p)$ against the number of top-scoring phenotypes included (mean and median across retained curves), the middle panel tracks how many SNPs’ curves are still running at each k — since early stopping ends curves at different points — and the bottom panel gives the distribution of peak k itself.

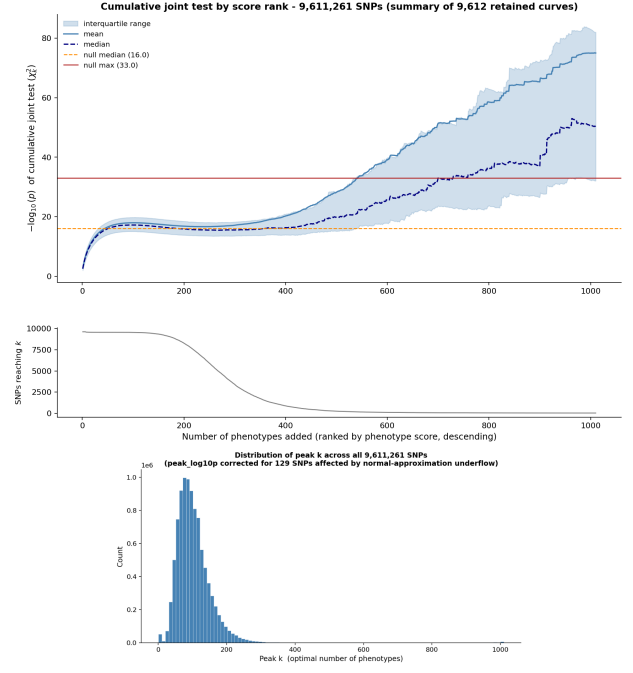
The two runs are similar in *shape* over the first few hundred phenotypes but diverge sharply beyond that. Both rise steeply over the first few dozen phenotypes and turn over near $k \approx 80$, and up to $k \approx 300$ the real curves sit only slightly above the null. Every one of the 9,611,261 null SNPs triggers the stopping rule where the last at $k = 552$ and no null curve ever recovers once it has turned over, consistent with a curve built from pure selection noise. The real curves, by contrast, climb again from $k \approx 400$ onward, crossing the null’s own maximum of 33.05 at around $k \approx 530$ and continuing to roughly 75 by $k = 1,010$. This late upturn is the signature the null calibration is designed to catch: phenotypes ranked far down the list are still contributing genuine association rather than merely adding degrees of freedom, which cannot happen when the underlying z-vector is pure noise. It is also why 0.4% of real SNPs never trigger the stopping rule at all and run the full length of the phenotype pool.

	median	p95	p99	max	peak k (median)
Null (simulated)	16.03	20.84	23.05	33.05	76
Real SNPs	18.80	27.08	38.22	1881.25*	94

Table 3: Peak $-\log_{10}(p)$ of the cumulative joint test across 9,611,261 real SNPs and 9,611,261 simulated null SNPs. *This peak occurs at $k = 1007$ — essentially the full joint test — and reflects amplification through a low-eigenvalue direction of $\hat{\Sigma}^{-1}$ rather than a uniquely strong association: no phenotype for this SNP reaches genome-wide significance ($\max |z| = 3.35$), addressed further below.



(a) Simulated null SNPs, $\mathbf{z} \sim \mathcal{N}(0, \hat{\Sigma}_{1010})$.



(b) 9,611,261 real SNPs.

Figure 16: Cumulative joint test of score-ranked phenotypes, for (a) 9,611,261 simulated null SNPs and (b) 9,611,261 real SNPs. Top: $-\log_{10}(p)$ against the number of top-scoring phenotypes included, summarized as mean and median across the retained curves. Middle: number of SNPs whose curve was still running at each k , since early stopping ends curves at different points. Bottom: distribution of peak k .

Table 3 quantifies what Figure 16 shows visually. The bulk of the real distribution sits inside the null’s own range — 95% of real SNPs peak below 27.08, against a null maximum of 33.05 so most of what the cumulative test appears to find in the real data is, at these percentiles, attributable to the selection procedure rather than genuine association. The tail tells a different story: the real 99th percentile (38.22) already exceeds the null maximum outright, and the real distribution continues well past it, reaching 90.62 at the 99.9th percentile against a null 99.9th percentile of only 27.98. Converting each real peak to an empirical p-value, 173,033 SNPs (1.80%) exceed the maximum observed across all 9,611,261 null draws, against an expectation of roughly one such SNP by chance alone, and since the selection procedure never once produced a peak above 33.05 in 9,611,261 attempts, these 173,033 SNPs cannot be explained by the ranking bias the null calibration was built to isolate.

A separate concern is numerical rather than statistical: for a subset of size k , the joint statistic is $S = \mathbf{z}_k^\top \Sigma_k^{-1} \mathbf{z}_k$, and when Σ_k is near-singular this can be inflated by directions of near-zero eigenvalue rather than by broad coordinated signal across phenotypes. We checked this directly for all 173,033 hits by eigendecomposition (`scan_eigenvalue_contamination.py`) of each SNP’s peak-subset covariance matrix and computing the fraction of S attributable to eigen-directions with $\lambda < 10^{-2}$ (explained below in **Interpretation**). The distribution is heavily concentrated at zero (median 0.000, 90th percentile 0.011, 99th percentile 0.062), confirming that degeneracy is not a broad driver of the discovery set. However, 26 SNPs (0.015%) have more than half their statistic explained this way and are flagged as artifacts rather than genuine coordinated effects. A few large- k ($k > 950$) cases where several dozen near-zero directions among hundreds dominate the statistic. Excluding these 26, the clean discovery count is **173,007 (1.8%)**.

The peak- k distribution (Figure 16, bottom panels) is less immediately informative than its height: the real median is 94 and mean 103.1, against a null median of 76 and mean 80.7, with the two distributions overlapping substantially. The concentration of peak k around 100 phenotypes therefore looks largely like a property of ranking 1,010 correlated phenotypes and taking a maximum, rather than a genuine feature of the association signal itself. It does, however, carry one concrete consequence for the sampling design of Section 7.2: a median peak k of 94 exceeds the upper bound of the subset size range used throughout this report, where k is drawn uniformly from $\{5, \dots, 90\}$. For more than half of all SNPs, the subset size giving the strongest cumulative signal is larger than any subset the random sampling procedure ever draws. This was a more specific limitation than the general coverage argument raised in Section 3.1 and again in Section 7.5, since the sampled subsets here are not merely a small fraction of the combinatorial space but are systematically undersized relative to where the real signal peaks.

It is worth flagging immediately that this count is of SNPs, not of independent genetic signals: unlike the null draws, which are simulated independently of one another, the real SNPs are correlated through LD (Section 1.2), so a single true underlying association can produce many nearby SNPs that all cross the null threshold together. The 173,007 figure is therefore better read as an upper bound on the number of independent discoveries, not a count of them; this is addressed further below.

Interpretation. Not all 173,007 SNPs represent genuine subset-based discoveries in the sense motivating this report. Of these, 10,073 (5.8%) peak at $k > 950$, where the cumulative test is effectively identical to the full joint test over all 1,010 phenotypes and whose significance is already captured by the joint column of Section 8.3; these are not subset findings at all. Direct inspection of the two most extreme cases in this region, **rs3217089** and **rs76016924**, shows their inflated statistics are *not* driven by coordinated genetic signal: neither has any phenotype reaching genome-wide significance ($\max |z| = 3.35$ and 3.49 respectively), and their summed squared z-scores are only $2.19\times$ and $1.10\times$ the null expectation, yet their joint statistics are inflated $12.1\times$ and $3.5\times$.

The source of this inflation is the eigenstructure of $\hat{\Sigma}_k$. Writing its orthonormal eigenvectors as v_i with eigenvalues λ_i , each v_i is a weighted combination of the k phenotypes, and λ_i is the variance of the z-scores along that combination. Note that a small λ_i marks a direction in which the phenotypes are nearly redundant. Since $\hat{\Sigma}_k$ is symmetric, it admits the spectral decomposition $\hat{\Sigma}_k = \sum_{i=1}^k \lambda_i v_i v_i^\top$, and because $\{v_i\}$ form an orthonormal basis, the inverse shares the same eigenvectors with reciprocal eigenvalues,

$$\hat{\Sigma}_k^{-1} = \sum_{i=1}^k \frac{1}{\lambda_i} v_i v_i^\top.$$

Therefore, the statistic S separates into independent contributions:

$$S = z^\top \hat{\Sigma}_k^{-1} z = \sum_{i=1}^k \frac{(v_i^\top z)^2}{\lambda_i},$$

each with expectation 1 under the null. Dividing by λ_i is exactly the intended behavior of the omnibus test – signal along a usually-redundant direction is genuinely more surprising. However, it leaves the statistic acutely sensitive to how precisely the smallest eigenvalues have been estimated. For **rs3217089**, the single smallest-eigenvalue direction alone contributes 44.4% of the total statistic, and

the 25 directions with $\lambda < 10^{-2}$ together contribute 63.8%; for **rs76016924**, the corresponding figures are 33.9% and 45.5%. The projections themselves, $(v_i^\top z)^2 = 8.89$ and 3.27 for the two dominant directions, are entirely unremarkable and consistent with the χ_1^2 they follow under the null so it is division by a near-zero eigenvalue, not an unusual projection, that manufactures the extreme statistic. This is also why the diagonal ridge of 10^{-4} introduced earlier in this section is not merely a numerical convenience: by placing a floor under how small λ_i can become, it directly bounds how much any single eigendirection can inflate S , at the cost of a small, quantified bias in the null covariance.

Limitations. The empirical p-value cannot resolve below 1.04×10^{-7} , so all 173,033 SNPs above the null maximum receive the same nominal value regardless of how far above it they individually fall, and are reported here as a count rather than a ranked list. This floor sits above the conventional genome-wide threshold of 5×10^{-8} , so per-SNP genome-wide significance cannot strictly be claimed from this calibration alone. The more defensible statement is the expected false-positive count of approximately one across the entire dataset. The simulated null SNPs are drawn independently of one another, whereas the real SNPs include variants in LD, so the real dataset does not in fact constitute 9,611,261 independent observations. Hence, converting the 173,033 SNPs into an estimate of independent loci (e.g. via distance- or LD-based clumping, collapsing correlated SNPs in the same genomic region into a single representative signal) is left to future work. Finally, 37,756 SNP identifiers in the working matrix are shared across more than one row, consistent with multi-allelic sites where several alternate alleles carry the same rsID; each row is nonetheless treated as an independent variant throughout this analysis, and variants are referenced by row position rather than by identifier alone to avoid ambiguity.

8.6.1 Comparing P-values: Univariate, Joint, and Optimal Subset

Aim. Section 8.3 compared three per-SNP tests: univariate, joint (all 1,010 phenotypes), and a Bonferroni-corrected minimum over 10,000 randomly sampled subsets. This subsection repeats that comparison with the subset test replaced by the constructively-built **optimal subset** of Section 8.6: the score-ranked prefix at which the cumulative joint test peaks, evaluated for every one of the 9,611,261 SNPs and assembled with the univariate and joint columns into a single output table (`final_output_table.parquet`) with shape 9,611,261 rows by 1,014 columns: `SNP_ID`, one phenotype-score column per trait (Section 8.1), `n_optimal_phenotypes`, `neglog10p_optimal_set`, and `neglog10p_joint_all_phenotypes`. All three source files feeding this table share identical row order with `pvalue_matrix.npy` (verified positionally against a common SNP-ID column), so the univariate column from Section 8.3 can be joined in directly.

Significance criterion. Unlike the univariate and joint columns, `neglog10p_optimal_set` is not a valid p-value on its own: phenotypes are ranked using the same z-scores that then enter the test, so its nominal chi-squared tail is inflated by selection (Section 8.6). Applying the standard $p < 5 \times 10^{-8}$ threshold to it directly flags 99.48% of all SNPs. This is an artifact of the ranking, not a result. Optimal-subset significance is therefore defined exactly as in Section 8.6: `neglog10p_optimal_set` exceeding 33.05, the largest peak produced by any of the 9,611,261 null draws under the identical selection procedure. Univariate and joint retain the standard $p < 5 \times 10^{-8}$ threshold, unchanged from Section 8.3.

Results. Table 4 breaks all 9,611,261 SNPs down by every combination of significance across the

three tests.

Category	SNPs (% of total)
None significant	7,666,428 (79.7651%)
Univariate only	4,776 (0.0497%)
Joint only	1,747,185 (18.1785%)
Optimal subset only	510 (0.0053%)
Univariate + Joint	19,839 (0.2064%)
Univariate + Optimal subset	22 (0.0002%)
Joint + Optimal subset	114,817 (1.1946%)
All three	57,684 (0.6002%)

Table 4: Contingency table of significance across the three tests, out of 9,611,261 SNPs.

Of the 173,033 SNPs that clear the calibrated optimal-subset threshold (Section 8.6), all but 532 are also significant under the joint test at $p < 5 \times 10^{-8}$: that is, $173,033 - 532 = 172,501$ (99.69%) of optimal-subset hits are also joint-significant. In other words, the optimal subset mostly sharpens a signal the joint test already detects rather than finding new territory. This is expected, since the calibrated threshold is a considerably stricter bar than the joint test’s fixed nominal cutoff, so clearing it in the first place already implies strong association.

The remaining 532 SNPs (0.31% of optimal-subset hits) are significant only under the optimal-subset test and not the joint test; of these, 510 are not univariate-significant either, so together these make up the **optimal subset only** and **univariate + optimal subset** categories in Table 4. Three of these 532 SNPs are among the 26 SNPs flagged above as eigenvalue-degenerate artifacts, leaving **529 as genuine subset-specific discoveries**.

This is directly comparable to Section 8.5’s finding from the random-sampling-based subset test: 666 subset-specific SNPs (259 subsets-only plus 407 univariate+subsets), against 532 here (510 subset-only plus 22 univariate+subset). The two subset constructions, random sampling over $k \in \{5, \dots, 90\}$ and constructive score-ranking up to $k = 1,010$, are built completely differently and draw from largely non-overlapping regions of the combinatorial space, and yet they arrive at a comparable order of magnitude of joint-invisible discoveries. This cross-validates that subset-based testing is finding a real, if narrow, category of signal, rather than an artifact of either particular construction method.

`n_optimal_phenotypes` differs systematically between the two groups: subset-specific SNPs (joint test misses them) peak at a median $k = 94$, versus a median $k = 133$ for SNPs significant under both the optimal subset and the joint test. This is consistent with Section 8.6’s own observation that the median peak k (94) already exceeds the $k \leq 90$ ceiling of the random-sampling procedure. The SNPs the joint test *can* still catch tend to need more phenotypes pooled together before the signal becomes overwhelming, while the truly subset-specific cases concentrate their signal in a comparatively smaller, more targeted prefix that the full 1,010-phenotype join dilutes away.

Figure 17 plots the joint and optimal-subset $-\log_{10}(p)$ against the univariate $-\log_{10}(p)$ for a representative sample of SNPs plus every optimal-subset hit. Both series sit almost entirely above the $y = x$ diagonal, confirming that combining phenotypes, whether via the full joint test or the constructed optimal subset, is essentially always at least as powerful as the best single phenotype. The optimal-subset series (red) tracks slightly above the joint series (blue) at a given univariate value across most

of the range, but for 40,399 SNPs (0.42%) the reverse holds. This is not a sign that the optimal subset was misidentified: `n_optimal_phenotypes` records only the k at which the early-stopping rule of Section 8.6 fired, not the k at which the true maximum occurs. For 78% of these 40,399 SNPs (`n_optimal_phenotypes` < 10, median 3), the cumulative curve peaks within the first few score-ranked phenotypes and then declines more than 30% from that peak, at which point the search stops; it never evaluates $k = 1,010$ at all. The joint test, computed independently over the full phenotype set rather than as a truncated search, can and does find signal these curves never got the chance to look for. This is precisely the failure mode Section 8.6 flags explicitly, that early stopping “returns the first local maximum whenever a curve declines and only later recovers, rather than the true global maximum.” The gap (`neglog10p_joint_all_phenotypes` – `neglog10p_optimal_set`) this produces is substantial where it occurs: median 2.10, reaching as high as $1,085 - \log_{10}(p)$ units. A separate, much smaller group of 1,360 SNPs (3.4% of the 40,399) behaves differently. These have `n_optimal_phenotypes` = 1,010 exactly, meaning their curves never triggered early stopping and ran to the full phenotype pool so the optimal subset genuinely is the full phenotype subset. For this group, the gap is small and tight, with a median of 0.86, and it always points in the same direction: optimal below joint, never above. This is the signature of the ridge term $10^{-4}\mathbf{I}$, added to $\hat{\Sigma}$ throughout the incremental Cholesky pipeline of Section 8.6. Remember that the joint test (Section 8.3) uses the plain, unregularized $\hat{\Sigma}$ instead. At matched $k = 1,010$, this ridge term systematically shrinks the optimal-subset statistic relative to the joint statistic, by a small and consistent amount.

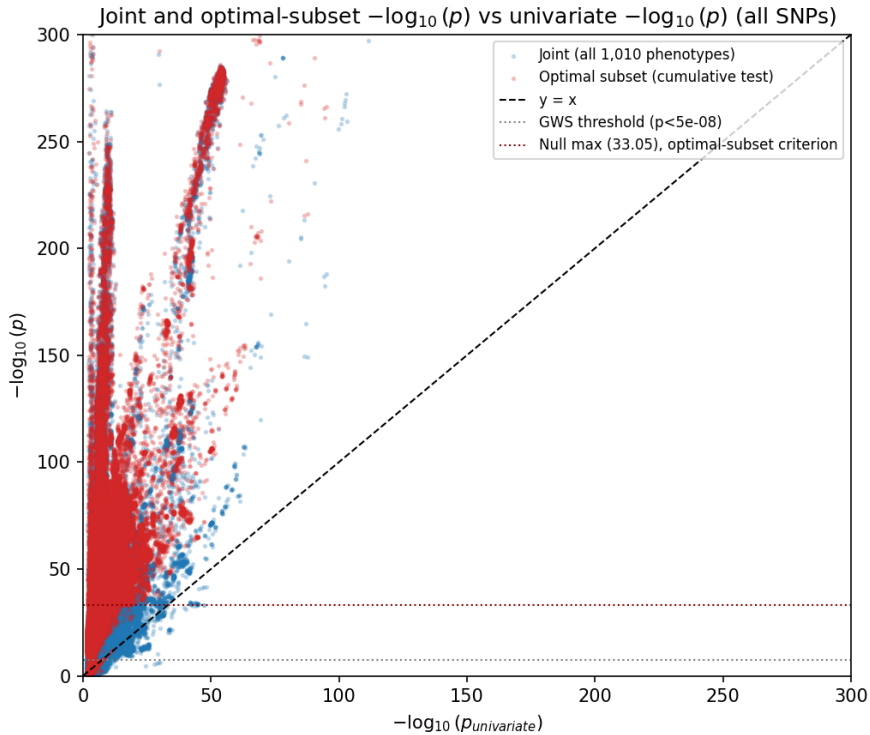


Figure 17: Q–Q-style comparison: $-\log_{10}(p)$ of the joint and optimal-subset tests plotted per SNP against the minimum univariate $-\log_{10}(p)$, restricted to $[0, 300]$ for readability. Axes clipped; a small number of SNPs exceed this range on all three tests. The null-max line (dark red) is the calibrated significance threshold for the optimal-subset test (Section 8.6); the GWS line (gray) is the standard threshold used for univariate and joint.

9 Conclusion and Next Steps

This report has followed a single method across five progressively larger scales: random subset sampling for multitrait GWAS testing, taken from a 31-subset proof of concept over five phenotypes (Section 3) to a full run over all 9,611,261 SNPs and 1,010 phenotypes (Section 7), followed by downstream analysis of that fullscale output (Section 8). At each stage, the pipeline’s correctness was checked against a naive, non-optimized reference before being trusted at the next, larger scale. This surfaced a genuine numerical bug (Sections 6.3, 7.3) that would otherwise have silently corrupted the final results.

The central empirical finding of Section 8 is two-sided, depending on how the subset test is used. Taken at the level of whole SNPs (Section 8.3), subset sampling’s unique contribution is narrow: only 259 of 9,611,261 SNPs reach significance exclusively through the subset test (Table 1), and of the 147,599 subset-significant SNPs, just **666** are missed by the joint test specifically. Where the subset test does contribute at this level, however, it does so in a way neither of the other two tests can: by surfacing signal carried by a specific, targeted combination of phenotypes rather than either one trait alone or all 1,010 at once (Section 8.4). Attributing that signal back to individual phenotypes proved harder than anticipated (Section 8.5): the average-score and winning-subset rankings do not agree, and no single phenotype dominates the 666 subset-specific SNPs the way one phenotype dominated the single-SNP case examined in Section 5.5.

Constructing the subset directly, rather than describing it after the fact, tells a meaningful story. The cumulative joint test of Section 8.6, ranking phenotypes by score and testing nested prefixes of that ranking against a matched-size simulated null, identifies **173,033 SNPs (1.80%)** whose peak association exceeds the largest peak produced across 9,611,261 null draws under the same selection procedure, against an expectation of roughly one such SNP by chance alone. This is an order of magnitude larger than the 666-SNP count because it asks a different question: not whether 10,000 random subsets happened to produce a stronger signal the joint test missed, but whether the *best achievable* score-ranked prefix carries signal the ranking bias itself cannot explain. The gap reflects search efficiency rather than a disagreement about the underlying truth: blind sampling explores a vanishing fraction of a $\approx 10^{304}$ -sized combinatorial space, while the cumulative test uses each SNP’s own phenotype score to walk directly toward the region most likely to carry signal for that SNP. The null calibration is what keeps this claim defensible rather than circular, since ranking phenotypes on the same data that then enters the test would otherwise inflate significance by construction.

Section 8.6.1 cross-validates this finding directly rather than only arguing for it on grounds of search efficiency, repeating the whole-SNP comparison of Section 8.3 with the optimal, score-ranked subset in place of the randomly sampled one. Of the 173,033 SNPs clearing the calibrated threshold, 532 (0.31%) are significant under the optimal subset but not the joint test, three of which coincide with eigenvalue-degenerate artifacts, leaving **529 genuine subset-specific discoveries**. This is comparable in order of magnitude to the 666-SNP count above, despite the two subset constructions being built completely differently and drawing from largely non-overlapping regions of the combinatorial space; two independent procedures converging on a similar count is stronger evidence of a real, if narrow, category of signal than either count alone. The two groups also differ consistently with the subset-size argument made above: subset-specific SNPs peak at a median $k = 94$, against $k = 133$ for SNPs the joint test also catches, so the signal unique to subset testing tends to concentrate in a smaller prefix than the full 1,010-phenotype join dilutes away.

This finding comes with two important qualifications, both surfaced in Section 8.6. First, not all 173,033 SNPs are subset discoveries in the sense motivating this report: 10,073 peak at $k > 950$, where the cumulative test is effectively the full joint test and offers nothing beyond Section 8.3. Inspection of the two most extreme such cases showed the apparent signal was manufactured by division through a near-zero eigenvalue of $\hat{\Sigma}_k$ rather than coordinated genetic effect, a failure mode worth bearing in mind wherever the omnibus statistic is applied to highly correlated phenotype pools. This is not an isolated occurrence: Section 8.6.1 identifies **26 SNPs** with the same eigenvalue-degeneracy pattern across the full dataset (`scan_eigenvalue_contamination.py`), three of which fall among the 532 apparent subset-specific discoveries above. Second, the median peak k across real SNPs (94) exceeds the upper bound of the subset size range used throughout sampling ($k \sim \text{Uniform}\{5, \dots, 90\}$): for more than half of all SNPs, the subset size carrying the strongest cumulative signal is larger than any subset the random sampling procedure was ever able to draw.

Reflecting on the project so far, a good amount of the effort on the fullscale run went into engineering rather than statistics: keeping peak memory under allocation via SNP-batching and memory-mapping (Section 7.2), tracking down float32 underflow in the survival function (Sections 6.3, 7.3), and verifying each optimization against a slower but transparent reference before trusting it. This progressive validation approach is the main methodological takeaway of the report, independent of the brain-MRI application: at genome-wide scale, a numerically “obviously correct” implementation is not the same as one that has actually been checked. A cheap, targeted verification (a handful of subsets, a handful of SNPs) catches the bulk of what fullscale verification would.

Beyond the engineering, this study demonstrates something more substantive: despite never covering the full combinatorial space of possible phenotype subsets, random subset sampling reliably produces smaller phenotype combinations that carry a larger association signal than the full 1,010-phenotype set, and often a signal invisible to univariate testing of any single phenotype in that combination. This is the core justification for the whole approach: a SNP whose effect is too weak or too diffuse to register in any one phenotype, and too diluted to register once all 1,010 are pooled together, can still be found once the right, smaller combination is tested jointly. What this study does not resolve is which phenotypes, or which combinations of them, are responsible for pushing signal up or down across SNPs in any generalizable way (Section 8.5); with an effectively endless number of combinations to search, finding such a pattern remains an open problem. This is the clearest argument for further research into exploring the combinatorial space of phenotype subsets more efficiently, without the sacrifice of extensive computational resources and runtime.

Next Steps to consider: Below, we will discuss some perspectives for further continuation of this research.

- **Revisiting the subset size range.** Re-running the sampling loop with a wider size range, informed by the peak- k distribution already in hand, would test whether a larger N_s and subset-size ceiling recovers additional signal in the 259 subsets-only SNPs of Section 8.5, or shifts the 173,033-SNP count from Section 8.6.
- **Converting SNP counts to independent loci.** Figure 16 from Section 8.6 counts correlated SNPs, not independent genetic signals, since real SNPs are linked through LD. Applying a clumping procedure such as collapsing SNPs within a genomic window (e.g. 250 kb–500 kb) or above a pairwise LD threshold into a single representative locus. This would convert this into an

effective count of independent discoveries, and this number matters for claims about how many distinct genetic signals the subset-based procedure is finding.

- **Missing data.** Rather than discarding SNPs with any missing z-score, implement a pipeline that filters per subset: a SNP is kept for a given subset only if all its phenotypes are observed, so a SNP is excluded just from the subsets it cannot support. Subset counts N_s scale with matrix sparsity (between $1.1 \times -1.5 \times$ baseline according to average-SNP sparsity).
- **Consolidation.** Complete derivation of LDSC intercept and genetic-correlation (standard error) Python script, and merge the scripts used at each stage into a single documented, parameterized pipeline (README.md, thread count as a configurable input, a toy dataset with expected output for testing).
- **Software development.** Discuss, with the unit's software engineers, packaging this pipeline into a shared, maintained tool rather than a sequence of per-stage notebooks.

References

- [1] Clare Bycroft, Colin Freeman, Desislava Petkova, Gavin Band, Lloyd T Elliott, Kevin Sharp, Allan Motyer, Damjan Vukcevic, Olivier Delaneau, Jared O’Connell, et al. The uk biobank resource with deep phenotyping and genomic data. *Nature*, 562(7726):203–209, 2018.
- [2] Peter M Visscher, Naomi R Wray, Qian Zhang, Pamela Sklar, Mark I McCarthy, Matthew A Brown, and Jian Yang. 10 years of gwas discovery: biology, function, and translation. *The American journal of human genetics*, 101(1):5–22, 2017.
- [3] Mahalanobis Pc. On the generalized distance in statistics. In *Proc Natl Inst Sci India*, volume 12, pages 49–55, 1936.
- [4] Gene H Golub and Charles F Van Loan. *Matrix computations*. JHU press, 2013.
- [5] Hanna Julienne, Pierre Lechat, Vincent Guillemot, Carla Lasry, Chunzi Yao, Robinson Araud, Vincent Laville, Bjarni Vilhjalmsen, Hervé Ménager, and Hugues Aschard. Jass: command line and web interface for the joint analysis of gwas results. *NAR genomics and bioinformatics*, 2(1): lqaa003, 2020.
- [6] Brendan K Bulik-Sullivan, Po-Ru Loh, Hilary K Finucane, Stephan Ripke, Jian Yang, Nick Patterson, Mark J Daly, Alkes L Price, and Benjamin M Neale. Ld score regression distinguishes confounding from polygenicity in genome-wide association studies. *Nature genetics*, 47(3):291–295, 2015.
- [7] Olive Jean Dunn. Multiple comparisons among means. *Journal of the American statistical association*, 56(293):52–64, 1961.
- [8] Brendan Bulik-Sullivan, Hilary K Finucane, Verner Anttila, Alexander Gusev, Felix R Day, Po-Ru Loh, ReproGen Consortium, Psychiatric Genomics Consortium, Genetic Consortium for Anorexia Nervosa of the Wellcome Trust Case Control Consortium 3, Laramie Duncan, et al. An atlas of genetic correlations across human diseases and traits. *Nature genetics*, 47(11):1236–1241, 2015.

10 Appendix

10.1 Claim: LDSC intercept matrix is an unbiased estimator of the null covariance matrix

We show that the null covariance of z-scores across two phenotypes i and k at the same SNP arises entirely from sample overlap and phenotypic correlation (intercept term), not from genetics. This justifies using the LDSC intercept matrix $\Sigma_{\text{int}} = \hat{\Sigma}_{\mathbf{K}}$ as the null covariance estimator in the omnibus statistic $\mathbf{S} = \mathbf{z}^\top \hat{\Sigma}_{\mathbf{K}}^{-1} \mathbf{z}$.

Proof. Consider K phenotypes and M SNPs. The estimated effect of SNP j on phenotype k is:

$$\hat{\beta}_{j,k} = \frac{\mathbf{x}_j^\top \mathbf{y}_k}{\mathbf{N}_{\mathbf{k}}}$$

where $\mathbf{x}_j \in \mathbb{R}^{N_k}$ is the standardized genotype vector, $\mathbf{y}_k \in \mathbb{R}^{N_k}$ is the standardized phenotype and $\mathbf{N}_{\mathbf{k}}$ denote the number of individuals genotyped and phenotyped for trait k .

The statistical model is that we assume the following simple linear model for phenotype k :

$$\mathbf{y}_k = \mathbf{X}\boldsymbol{\beta}_k + \boldsymbol{\varepsilon}_k$$

where $\mathbf{X} \in \mathbb{R}^{N_k \times M}$, $\boldsymbol{\beta}_k \in \mathbb{R}^M$ is the vector of true causal effect sizes of all M SNPs on phenotype k , and $\boldsymbol{\varepsilon}_k \in \mathbb{R}^{N_k} \sim \mathcal{N}(0, \sigma_k^2 \mathbf{I})$ is the vector of residuals errors across all N_k individuals, assumed to be independent of $\mathbf{X}$ and capturing variation in the phenotype NOT explained by genetics.

The OLS estimator for $\beta_{j,k}$ from the marginal regression that considers a single SNP j , replacing $\mathbf{X}$ with $\mathbf{x}_j$ and $\boldsymbol{\beta}_k$ with the scalar $\beta_{j,k}$: $\mathbf{y}_k = \mathbf{x}_j \beta_{j,k} + \boldsymbol{\varepsilon}_k$ is:

$$\hat{\beta}_{j,k} = (\mathbf{x}_j^\top \mathbf{x}_j)^{-1} \mathbf{x}_j^\top \mathbf{y}_k.$$

Since $\boldsymbol{\varepsilon}_k \sim \mathcal{N}(\mathbf{0}, \sigma_k^2 \mathbf{I})$ and $\mathbf{x}_j$ is treated as fixed, the variance of this estimator is

$$\begin{aligned} \text{Var}(\hat{\beta}_{j,k}) &= (\mathbf{x}_j^\top \mathbf{x}_j)^{-1} \mathbf{x}_j^\top \text{Var}(\mathbf{y}_k) \mathbf{x}_j (\mathbf{x}_j^\top \mathbf{x}_j)^{-1} \\ &= (\mathbf{x}_j^\top \mathbf{x}_j)^{-1} \mathbf{x}_j^\top (\sigma_k^2 \mathbf{I}) \mathbf{x}_j (\mathbf{x}_j^\top \mathbf{x}_j)^{-1} \\ &= \sigma_k^2 (\mathbf{x}_j^\top \mathbf{x}_j)^{-1}, \end{aligned}$$

and therefore

$$\text{SE}(\hat{\beta}_{j,k}) = \frac{\sigma_k}{\sqrt{\mathbf{x}_j^\top \mathbf{x}_j}}.$$

Since $\mathbf{x}_j$ is standardized (mean zero, variance one), $\mathbf{x}_j^\top \mathbf{x}_j = N_k$. Furthermore, under the polygenic assumption each SNP explains a negligible fraction of phenotypic variance ($\beta_{j,k}^2 \approx 0$, the true effect), so the residual variance of phenotype k satisfies $\sigma_k^2 \approx 1$. Combining these two approximations gives

$\text{SE}(\hat{\beta}_{j,k}) \approx 1/\sqrt{N_k}$, and the z-score reduces to

$$z_{j,k} = \frac{\hat{\beta}_{j,k}}{\text{SE}(\hat{\beta}_{j,k})} \approx \sqrt{N_k} \cdot \hat{\beta}_{j,k}.$$

Having established $z_{j,k} \approx \sqrt{N_k} \cdot \hat{\beta}_{j,k}$, we now derive the null covariance structure of z-scores *across phenotypes* at the same SNP j . The (i, k) entry of the null covariance matrix $\hat{\Sigma}_K$ is precisely $\mathbb{E}[z_{j,i} z_{j,k}]$.

Decomposition of $\mathbb{E}[z_{j,i} z_{j,k}]$. This is measuring the expected correlation between the z-scores of two different phenotypes i and k at the same SNP j , averaged over all null SNPs in the genome. That is, how much the z-scores of two phenotypes co-move at the same SNP purely due to study design, not due to genetics.

Substituting the statistical model:

$$\hat{\beta}_{j,k} = \frac{\mathbf{x}_j^\top \mathbf{X} \boldsymbol{\beta}_k}{N_k} + \frac{\mathbf{x}_j^\top \boldsymbol{\varepsilon}_k}{N_k} = \underbrace{\sum_l r_{j,l} \beta_{l,k}}_{\text{genetic signal}} + \underbrace{\frac{\mathbf{x}_j^\top \boldsymbol{\varepsilon}_k}{N_k}}_{\text{noise (intercept)}}$$

where $r_{j,l} = \frac{\mathbf{x}_j^\top \mathbf{x}_l}{N_k}$ is the sample correlation between the standardized genotypes at SNPs j and l (the LD correlation). Therefore, noting that $z_{j,k} \approx \sqrt{N_k} \cdot \hat{\beta}_{j,k}$:

$$\mathbb{E}[z_{j,i} z_{j,k}] = \underbrace{\sqrt{N_i N_k} \sum_{l,m} r_{j,l} r_{j,m} \mathbb{E}[\beta_{l,i} \beta_{m,k}]}_{\text{genetic term}} + \underbrace{\frac{\mathbb{E}[\mathbf{x}_j^\top \boldsymbol{\varepsilon}_i \cdot \mathbf{x}_j^\top \boldsymbol{\varepsilon}_k]}{\sqrt{N_i N_k}}}_{\text{intercept term}}$$

The two mixed (genetic $\times$ noise) cross-terms vanish in expectation, since x_j and the residuals ε are independent with $E[\varepsilon] = 0$. We first focus on the intercept term, which is not contaminated by any genetic signal.

Null (intercept) term

To derive the intercept term, we decompose each phenotype into a genetic signal and an environmental residual: $\mathbf{y}_k = \mathbf{X} \boldsymbol{\beta}_k + \boldsymbol{\varepsilon}_k$, where $\boldsymbol{\varepsilon}_k$ is the vector of residuals across all N_k individuals. For a null SNP j (no causal effect), the genetic signal contributes nothing to $z_{j,i} z_{j,k}$, and what remains is the cross-product of residuals.

Substituting $z_{j,k} \approx x_j^\top y_k / \sqrt{N_k}$ and the linear model $y_k = X \boldsymbol{\beta}_k + \varepsilon_k$, the product of z-scores expands into four terms:

$$\begin{aligned} z_{j,i} z_{j,k} &= \frac{x_j^\top y_i}{\sqrt{N_i}} \cdot \frac{x_j^\top y_k}{\sqrt{N_k}} = \frac{x_j^\top (X \boldsymbol{\beta}_i + \boldsymbol{\varepsilon}_i)}{\sqrt{N_i}} \cdot \frac{x_j^\top (X \boldsymbol{\beta}_k + \boldsymbol{\varepsilon}_k)}{\sqrt{N_k}} \\ &= \frac{1}{\sqrt{N_i N_k}} \left[\underbrace{(x_j^\top X \boldsymbol{\beta}_i)(x_j^\top X \boldsymbol{\beta}_k)}_{\text{genetic} \rightarrow \ell_j[\Sigma_g]_{i,k}} + \underbrace{(x_j^\top X \boldsymbol{\beta}_i)(x_j^\top \boldsymbol{\varepsilon}_k) + (x_j^\top \boldsymbol{\varepsilon}_i)(x_j^\top X \boldsymbol{\beta}_k)}_{\text{cross terms} \rightarrow 0} + \underbrace{(x_j^\top \boldsymbol{\varepsilon}_i)(x_j^\top \boldsymbol{\varepsilon}_k)}_{\text{intercept (noise)}} \right]. \quad (1) \end{aligned}$$

Nothing is discarded by force here: at a null SNP the genetic contribution $x_j^\top X \boldsymbol{\beta}$ does not vanish, but

is precisely the slope term $\ell_j \Sigma_g$ derived separately in the **Genetic signal term** subsection below. Taking the expectation, the two cross terms vanish because effect sizes and residuals are independent with $\mathbb{E}[\varepsilon] = 0$, e.g. $\mathbb{E}[(x_j^\top X \beta_i)(x_j^\top \varepsilon_k)] = \mathbb{E}[x_j^\top X \beta_i \cdot x_j^\top \mathbb{E}[\varepsilon_k]] = 0$. This subsection evaluates only the remaining intercept (noise) term.

Isolating this intercept term, and since x_j is independent of the residuals and standardized ($\mathbb{E}[x_{j,n}^2] = 1$, $\mathbb{E}[x_{j,n}x_{j,m}] = 0$ for $n \neq m$), we have

$$\mathbb{E}[(x_j^\top \varepsilon_i)(x_j^\top \varepsilon_k)] = \sum_{n,m} \varepsilon_{i,n} \varepsilon_{k,m} \mathbb{E}[x_{j,n}x_{j,m}] = \sum_n \varepsilon_{i,n} \varepsilon_{k,n} = \varepsilon_i^\top \varepsilon_k,$$

where the cross terms vanish by independence across individuals and the diagonal reduces to 1 by unit variance. Note this differs from $\mathbb{E}[x_j^\top x_j] = N_k$, which is a sum of squared genotypes along the same vector. Here the two residual vectors ε_i and ε_k pair each individual's contributions, giving a simple inner product with no N_k prefactor. Hence the intercept entry of the null covariance matrix is

$$[\Sigma_{\text{int}}]_{i,k} = \frac{1}{\sqrt{N_i N_k}} \mathbb{E}[\varepsilon_i^\top \varepsilon_k].$$

This expectation is non-zero for two reasons. First, if $N_s^{(i,k)}$ i.e., number of shared individuals, were measured in both study i and study k (sample overlap), their residuals co-vary: for each shared individual n , $\mathbb{E}[\varepsilon_{i,n} \cdot \varepsilon_{k,n}] = \rho_{i,k}$, the phenotypic correlation between traits i and k .¹ Second, individuals measured in only one study contribute zero, since their residuals are independent. Summing over all individuals, only the $N_s^{(i,k)}$ shared ones contribute:

$$\mathbb{E}[\varepsilon_i^\top \varepsilon_k] = N_s^{(i,k)} \rho_{i,k}$$

which gives the intercept term:

$$\frac{N_s^{(i,k)} \rho_{i,k}}{\sqrt{N_i N_k}}$$

This term captures only sample overlap and phenotypic correlation, NOT genetics. Combined across all phenotype pairs, the identity holds for every entry (i, k) so it holds for the full matrix:

$$\boxed{\mathbb{E}[\mathbf{z}_j \mathbf{z}_j^\top] = \ell_j \Sigma_g + \Sigma_{\text{int}}}$$

where $\mathbf{z}_j \in \mathbb{R}^K$, $[\Sigma_g]_{i,k}$ is the genetic signal to be defined, $[\Sigma_{\text{int}}]_{i,k} = \frac{N_s^{(i,k)} \rho_{i,k}}{\sqrt{N_i N_k}}$ (phenotypic correlation, sample overlap), and $\ell_j = \sum_{l \neq j} r_{j,l}^2$ is the LD score of SNP j , summed over all other SNPs $l \neq j$.

To compute the test statistic $S_j = \mathbf{z}_j^\top \hat{\Sigma}^{-1} \mathbf{z}_j$, we need an estimate of Σ_{int} that captures **only** sample overlap and phenotypic correlation. A natural candidate is the sample mean of $\mathbf{z}_j \mathbf{z}_j^\top$ over all SNPs:

¹Strictly, $\rho_{i,k}$ here is the correlation of the additively-unexplained (non-genetic) residuals $\varepsilon_i, \varepsilon_k$, not the raw phenotypic correlation of y_i, y_k ; the two coincide exactly only under the same negligible-per-SNP-heritability approximation ($\sigma_k^2 \approx 1$) already invoked above.

$$\hat{\Sigma}_{\text{emp}} = \frac{1}{M} \sum_j \mathbf{z}_j \mathbf{z}_j^\top$$

Taking expectations:

$$\mathbb{E}[\hat{\Sigma}_{\text{emp}}] = \frac{1}{M} \sum_j \mathbb{E}[\mathbf{z}_j \mathbf{z}_j^\top] = \frac{1}{M} \sum_j (\ell_j \Sigma_g + \Sigma_{\text{int}}) = \bar{\ell} \Sigma_g + \Sigma_{\text{int}}$$

where $\bar{\ell} = \frac{1}{M} \sum_j \ell_j > 0$ is the mean LD score. The bias is $\bar{\ell} \Sigma_g$, which inflates every off-diagonal entry by the average genetic covariance weighted by LD. Consequently $\hat{\Sigma}_{\text{emp}}$ is inflated relative to Σ_{int} , so the test statistic:

$$S_j = \mathbf{z}_j^\top \hat{\Sigma}_{\text{emp}}^{-1} \mathbf{z}_j$$

is deflated: the test is conservative meaning p-values too large which decreases risk of type 1 error (false positive). Moreover, it is underpowered meaning it increases the risk of type 2 errors (false negatives). So rather than using Σ_{emp} , we regress the observed z-score products $z_{j,i} z_{j,k}$ on the LD scores ℓ_j across all M SNPs and take the intercept, which is uncontaminated by the genetic term.

The identity $\mathbb{E}[z_{j,i} z_{j,k}] = \ell_j [\Sigma_g]_{i,k} + [\Sigma_{\text{int}}]_{i,k}$ is a linear regression of $z_{j,i} z_{j,k}$ on ℓ_j across SNPs. We derive this from the **LDSC regression**: a simple linear regression of the observed z-score products $z_{j,i} z_{j,k}$ on the LD score ℓ_j across all M SNPs, with slope $[\Sigma_g]_{i,k}$ and intercept $[\Sigma_{\text{int}}]_{i,k}$:

$$z_{j,i} z_{j,k} = \underbrace{[\Sigma_g]_{i,k}}_{\text{slope}} \ell_j + \underbrace{[\Sigma_{\text{int}}]_{i,k}}_{\text{intercept}} + \eta_j \quad (2)$$

where η_j is the **regression residual** for SNP j . We treat the residuals η_j as independent and identically distributed (i.i.d.) random variables with $\mathbb{E}[\eta_j] = 0$ (this is an approximation, as nearby SNPs in LD have correlated z-scores, but holds asymptotically as LD decays with genomic distance). By the Weak Law of Large Numbers, for an i.i.d. sequence $\eta_1, \eta_2, \dots, \eta_M$ with $\mathbb{E}[\eta_j] = 0$ for $j \in I_M$ where $M = 9,611,261$, the sample average converges to the expected value:

$$\bar{\eta}_M = \frac{1}{M} \sum_{j=1}^M \eta_j \xrightarrow{p} \mathbb{E}[\eta_j] = 0 \quad \text{as } M \rightarrow \infty$$

In the present dataset, while the convergence is asymptotic in theory, we assume M is large enough in practice that the noise contribution $\bar{\eta}_M$ is negligible, and the OLS estimates of the slope and intercept are dominated by the systematic linear signal. This means that each observed product $z_{j,i} z_{j,k}$ deviates from its expectation by η_j ; regressing across all M SNPs averages out this noise, and the intercept recovers $[\Sigma_{\text{int}}]_{i,k}$. This intercept is identified as the expected cross-product in the limit $\ell_j \rightarrow 0$, i.e., at a SNP with no LD to any other SNP which is a purely null contribution. The intercept estimator $\hat{\mathbf{c}}_{\mathbf{i},\mathbf{k}}$ of $[\Sigma_{\text{int}}]_{i,k}$ is defined as the OLS intercept from fitting regression (2) across all M SNPs.

$\hat{\mathbf{c}}_{\mathbf{i},\mathbf{k}}$ derivation. Denote $y_j = z_{j,i} z_{j,k}$, $f(x_j) = \alpha x_j + \beta$ and $x_j = \ell_j$ for brevity.

OLS minimizes the sum of squared residuals:

$$R(\alpha, \beta) = \sum_{j=1}^M (y_j - f(x_j))^2 = \sum_{j=1}^M (y_j - (\alpha x_j + \beta))^2 = \sum_{j=1}^M (y_j - \alpha x_j - \beta)^2$$

Taking partial derivatives and setting them to zero gives the normal equations:

$$\frac{\partial R}{\partial \beta} = 0 \implies -2 \sum_{j=1}^M (y_j - \alpha x_j - \beta) = 0 \quad (3)$$

$$\frac{\partial R}{\partial \alpha} = 0 \implies -2 \sum_{j=1}^M x_j (y_j - \alpha x_j - \beta) = 0 \quad (4)$$

From equation (3):

$$\sum_{j=1}^M y_j - \alpha \sum_{j=1}^M x_j - M\beta = 0 \implies \hat{\beta} = \bar{y} - \alpha \bar{x}$$

where $\bar{y} = \frac{1}{M} \sum_j y_j$ and $\bar{x} = \frac{1}{M} \sum_j x_j$. Substituting into equation (4):

$$\begin{aligned} \sum_{j=1}^M x_j (y_j - \alpha x_j - \bar{y} + \alpha \bar{x}) &= 0 \\ \sum_{j=1}^M x_j [(y_j - \bar{y}) - \alpha(x_j - \bar{x})] &= 0 \\ \sum_{j=1}^M x_j (y_j - \bar{y}) &= \alpha \sum_{j=1}^M x_j (x_j - \bar{x}) \end{aligned} \quad (5)$$

Expanding the left-hand side of equation (4). Write $x_j = (x_j - \bar{x}) + \bar{x}$:

$$\begin{aligned} \sum_{j=1}^M x_j (y_j - \bar{y}) &= \sum_{j=1}^M [(x_j - \bar{x}) + \bar{x}] (y_j - \bar{y}) \\ &= \sum_{j=1}^M (x_j - \bar{x})(y_j - \bar{y}) + \bar{x} \sum_{j=1}^M (y_j - \bar{y}) \end{aligned}$$

Since $\sum_{j=1}^M (y_j - \bar{y}) = \sum_{j=1}^M y_j - M\bar{y} = 0$:

$$\sum_{j=1}^M x_j (y_j - \bar{y}) = \sum_{j=1}^M (x_j - \bar{x})(y_j - \bar{y})$$

Similarly with $x_j = (x_j - \bar{x}) + \bar{x}$:

$$\begin{aligned} \sum_{j=1}^M x_j (x_j - \bar{x}) &= \sum_{j=1}^M [(x_j - \bar{x}) + \bar{x}] (x_j - \bar{x}) \\ &= \sum_{j=1}^M (x_j - \bar{x})^2 + \bar{x} \sum_{j=1}^M (x_j - \bar{x}) \end{aligned}$$

Since $\sum_{j=1}^M (x_j - \bar{x}) = 0$:

$$\sum_{j=1}^M x_j (x_j - \bar{x}) = \sum_{j=1}^M (x_j - \bar{x})^2$$

Substituting back:

$$\hat{\alpha}_{i,k} = \frac{\sum_{j=1}^M (x_j - \bar{x})(y_j - \bar{y})}{\sum_{j=1}^M (x_j - \bar{x})^2} = \frac{\sum_{j=1}^M (\ell_j - \bar{\ell})(z_{j,i} z_{j,k} - \overline{z_i z_k})}{\sum_{j=1}^M (\ell_j - \bar{\ell})^2} = \frac{\widehat{\text{Cov}}(\ell_j, z_{j,i} z_{j,k})}{\widehat{\text{Var}}(\ell_j)}$$

where $\overline{z_i z_k} = \frac{1}{M} \sum_j z_{j,i} z_{j,k}$ the mean product, $\bar{\ell} = \frac{1}{M} \sum_j \ell_j$ the mean LD score and $\hat{\alpha}_{i,k}$ is the OLS **slope (genetic signal) estimator**.

Therefore, the OLS **intercept estimator**:

$$\hat{c}_{i,k} = \bar{y} - \hat{\alpha}_{i,k} \bar{x} = \frac{1}{M} \sum_{j=1}^M z_{j,i} z_{j,k} - \hat{\alpha}_{i,k} \cdot \frac{1}{M} \sum_{j=1}^M \ell_j = \overline{z_i z_k} - \hat{\alpha}_{i,k} \bar{\ell}$$

Definition: Well-specified. Given a model $\{p_\theta | \theta \in \Theta\}$, then there exists a $\theta^* \in \Theta$ such that $p_{\theta^*} = p^*$, where p^* is the true data-generating distribution and Θ is the parameter space

For the LDSC regression, the model family is the set of linear models

$$z_{j,i} z_{j,k} = \alpha \ell_j + \beta + \eta_j, \quad (\alpha, \beta) \in \mathbb{R}^2 = \Theta$$

and the derivation of $\mathbb{E}[z_{j,i} z_{j,k}]$ shows that the true conditional expectation is exactly linear in ℓ_j with $\mathbb{E}[\eta_j] = 0$ by construction (see **Genetic term** below), so the true parameters $\theta^* = ([\Sigma_g]_{i,k}, [\Sigma_{\text{int}}]_{i,k})$ exist within Θ . The model is therefore well-specified, and since OLS is unbiased under a well-specified linear model with exogenous predictor (here $E[\eta_j | \ell_j] = 0$, strengthening the $E[\eta_j] = 0$ assumed above),

$$\mathbb{E}[\hat{c}_{i,k}] = [\Sigma_{\text{int}}]_{i,k}.$$

Assembling these entries over all pairs (i, k) , the LDSC intercept matrix $[\Sigma_{\text{int}}] = [\hat{c}_{i,k}]_{i,k}$ is therefore an unbiased estimator of the null covariance matrix $\Sigma_{\mathbf{K}}$, which is the matrix $\hat{\Sigma}_{\mathbf{K}}$ used as the null covariance in the omnibus statistic $\mathbf{S} = \mathbf{z}^\top \hat{\Sigma}_{\mathbf{K}}^{-1} \mathbf{z}$.

Specialization to a single cohort. The expression above is fully general: it allows the phenotypes to come from separate studies with different sample sizes N_i, N_k and partial overlap $N_s^{(i,k)}$. In this project, however, all K phenotypes are measured on the *same* UKB cohort, so every pair of traits shares all individuals:

$$N_i = N_k = N_s^{(i,k)} = N = 32,947.$$

The intercept term then collapses,

$$[\Sigma_{\text{int}}]_{i,k} = \frac{N_s^{(i,k)} \rho_{i,k}}{\sqrt{N_i N_k}} = \frac{N \rho_{i,k}}{\sqrt{N \cdot N}} = \rho_{i,k},$$

so that Σ_{int} is exactly the $K \times K$ **phenotypic correlation matrix** $[\rho_{i,k}]_{i,k}$ (with unit diagonal, since $\rho_{i,i} = 1$). This is the reduction invoked in Section 2.2: the sample-overlap coefficient $N_s^{(i,k)} / \sqrt{N_i N_k}$

equals 1 for every pair, leaving only the phenotypic correlation.

In practice, the diagonal entries of the estimated Σ_{int} deviate slightly from 1 due to estimation noise in the LDSC regression and residual confounding from population stratification. These deviations are small and negligible (typically $|\Sigma_{\text{int},i,i} - 1| < 0.05$) to the the null calibration of S .

Genetic signal term

Unlike the intercept term, the genetic term cannot be evaluated from the statistical model alone: it requires an explicit assumption on how causal effects are distributed across the genome. We adopt the infinitesimal polygenic model of Bulik-Sullivan et al. [8], generalized from two phenotypes to K , and specialized to the case relevant here where all K phenotypes are measured on the *same* cohort (a single genotype matrix $\mathbf{X}$, rather than two separate studies) which is precisely the "same study" reduction noted in that paper.

Assumption (polygenicity). For every SNP l , the true per-normalized-genotype effect sizes $(\beta_{l,1}, \dots, \beta_{l,K})$ across the K phenotypes are mean zero, $\mathbb{E}[\beta_{l,i}] = 0$ for every SNP l and phenotype i , and causal effects at different SNPs are independent. At any single SNP l , the $K \times K$ covariance of causal effects across phenotypes is the same for every SNP in the genome:

$$\text{Cov}(\beta_{l,i}, \beta_{l,k}) = \frac{\sigma_{i,k}}{M} \quad \text{for every SNP } l,$$

where $\sigma_{i,k}$ is the (i, k) entry of a $K \times K$ matrix of per-normalised-genotype genetic covariances (with $\sigma_{i,i} = h_i^2$, the SNP-heritability of phenotype i). This is the direct K -phenotype extension of $\text{Var}[(\beta, \gamma)] = \frac{1}{M} \begin{pmatrix} h_1^2 I & \rho_g I \\ \rho_g I & h_2^2 I \end{pmatrix}$ (the subscript g denoting *genetic* so ρ_g is genetic correlation): the I (identity matrix, indexed by SNP) is exactly the statement that different SNPs are independent, while ρ_g , a single number describing the genetic covariance between the two phenotypes in that model, generalizes to $\sigma_{i,k}$: the same quantity, but for an arbitrary pair (i, k) drawn from the pool of K phenotypes considered here. Note β and γ denote the causal-effect vectors of the first and second phenotype respectively.

Combining independence across SNPs with the homogeneous within-SNP covariance:

$$\text{Cov}(\beta_{l,i}, \beta_{m,k}) = \frac{\sigma_{i,k}}{M} \mathbf{1}_{l=m}. \quad (*)$$

Remark (independent causal effects vs. correlated genotypes). It may seem inconsistent to assume that causal effects are independent across SNPs while simultaneously relying on LD, i.e. correlated genotypes, elsewhere in the derivation. It is not: the model in Bulik-Sullivan et al. [6] specifies these as two separate, independently chosen pieces of the process, and the whole point of LD Score Regression rests on them being different.

Explicitly, their model places the correlation structure entirely on the *genotype* side: individuals' genotype vectors are drawn with covariance matrix r , i.e. $\text{Cov}(x_j, x_l) = r_{j,l}$, the LD between SNPs j and l . The *effect-size* prior is specified separately, as $\text{Var}[(\beta, \gamma)] = \frac{1}{M} \begin{pmatrix} h_1^2 I & \rho_g I \\ \rho_g I & h_2^2 I \end{pmatrix}$ and by construction, $\text{Cov}(\beta_l, \beta_m) = 0$ for $l \neq m$. Nothing in the model forces these two covariance structures to agree, because they answer different questions: one is a statement about population

genetics (how genotypes co-occur due to shared ancestry and recombination), the other about genetic architecture (how strongly each SNP truly influences the trait) . Hence a modeler is free to specify each independently.

Correlated genotypes and independent causal effects are not merely compatible. In fact, this combination is exactly the mechanism LD Score Regression depends on: correlated genotypes are what allow independent causal effects to produce correlated marginal estimates. Derived above, the marginal estimate at SNP j is a genotype-weighted sum over *every* SNP's true effect,

$$\hat{\beta}_{j,k} = \sum_l r_{j,l} \beta_{l,k} + \text{noise},$$

so even though $\beta_{l,k}$ and $\beta_{m,k}$ are independent for $l \neq m$, the marginal estimates $\hat{\beta}_{j,k}$ and $\hat{\beta}_{i,k}$ at two SNPs i, j in LD with a shared causal neighbour l both inherit a contribution from that same $\beta_{l,k}$, weighted by $r_{j,l}$ and $r_{i,l}$ respectively, and so become correlated. This is exactly how LD propagates signal from a causal SNP to its correlated neighbours. If causal effects were instead correlated in step with LD, the genetic term's double sum would pick up extra off-diagonal contributions, and the result would no longer be exactly proportional to ℓ_j .

This independence is an assumption about genetic architecture, not something we know to be true of real genomes. It is standard in the infinitesimal polygenic model underlying LD Score Regression, and known to be robust to moderate violations, though it can break down in extreme cases (e.g. if all causal variants are rare and clustered).

Step 1: Expectation of a product as covariance plus product of means. For any two random variables X, Y with finite variance, $\mathbb{E}[XY] = \text{Cov}(X, Y) + \mathbb{E}[X]\mathbb{E}[Y]$. Applying this with $X = \beta_{l,i}$, $Y = \beta_{m,k}$:

$$\mathbb{E}[\beta_{l,i}\beta_{m,k}] = \text{Cov}(\beta_{l,i}, \beta_{m,k}) + \mathbb{E}[\beta_{l,i}] \mathbb{E}[\beta_{m,k}].$$

By the mean-zero part of the assumption, $\mathbb{E}[\beta_{l,i}] = \mathbb{E}[\beta_{m,k}] = 0$, so the second term vanishes:

$$\mathbb{E}[\beta_{l,i}\beta_{m,k}] = \text{Cov}(\beta_{l,i}, \beta_{m,k}).$$

This step alone does not yet give a closed form as it only re-expresses an expectation as a covariance, which is still an $M \times M$ array of unknowns.

Step 2: Apply the polygenic covariance structure. Substituting (*):

$$\mathbb{E}[\beta_{l,i}\beta_{m,k}] = \frac{\sigma_{i,k}}{M} \mathbf{1}_{l=m}.$$

This is the key simplification: the expected product of causal effects is exactly zero whenever $l \neq m$ (different SNPs contribute independently to the sum below), and equal to the constant $\sigma_{i,k}/M$ whenever $l = m$, regardless of which SNP l is.

Step 3: Substitute into the genetic term and collapse the double sum. Recall the genetic term is $\sqrt{N_i N_k} \sum_{l,m} r_{j,l} r_{j,m} \mathbb{E}[\beta_{l,i}\beta_{m,k}]$. Substituting Step 2, every term with $l \neq m$ vanishes, leaving

only the diagonal $l = m$:

$$\begin{aligned}\sqrt{N_i N_k} \sum_{l,m} r_{j,l} r_{j,m} \mathbb{E}[\beta_{l,i} \beta_{m,k}] &= \sqrt{N_i N_k} \sum_{l,m} r_{j,l} r_{j,m} \frac{\sigma_{i,k}}{M} \mathbf{1}_{l=m} \\ &= \frac{\sqrt{N_i N_k} \sigma_{i,k}}{M} \sum_l r_{j,l}^2.\end{aligned}$$

Step 4: Identify the LD score. As mentioned, the LD score of SNP j is $\ell_j = \sum_l r_{j,l}^2$, summed over all *other* SNPs $l \neq j$. The sum $\sum_l r_{j,l}^2$ above technically runs over all M SNPs including $l = j$ itself, where $r_{j,j} = 1$ contributes exactly 1 to the sum. Since ℓ_j is typically on the order of tens to hundreds of correlated neighbours genome-wide, this single self-term is negligible. Hence

$$\sum_l r_{j,l}^2 \approx \ell_j.$$

Step 5: Closed-form expression of genetic term. Combining Steps 3–4:

$$\sqrt{N_i N_k} \sum_{l,m} r_{j,l} r_{j,m} \mathbb{E}[\beta_{l,i} \beta_{m,k}] = \frac{\sqrt{N_i N_k} \sigma_{i,k}}{M} \ell_j.$$

This is exactly the coefficient of ℓ_j identified above, $[\Sigma_g]_{i,k} = \frac{\sqrt{N_i N_k} \sigma_{i,k}}{M}$: the genetic term is linear in the LD score ℓ_j , with slope $[\Sigma_g]_{i,k}$ and no dependence on which specific SNP j is being considered beyond its LD score. This is precisely the structure that makes LD Score Regression **well-specified**.

Conclusion

Therefore combining both the genetic and intercept term we yield:

$$\begin{aligned}\mathbb{E}[z_{j,i} z_{j,k}] &= \underbrace{\sqrt{N_i N_k} \sum_{l,m} r_{j,l} r_{j,m} \mathbb{E}[\beta_{l,i} \beta_{m,k}]}_{\text{genetic term}} + \underbrace{\frac{\mathbb{E}[\mathbf{x}_j^\top \boldsymbol{\varepsilon}_i \cdot \mathbf{x}_j^\top \boldsymbol{\varepsilon}_k]}{\sqrt{N_i N_k}}}_{\text{intercept term}} \\ &= \frac{\sqrt{N_i N_k} \sigma_{i,k}}{M} \ell_j + \frac{N_s^{(i,k)} \rho_{i,k}}{\sqrt{N_i N_k}} \\ &= [\Sigma_g]_{i,k} \ell_j + [\Sigma_{int}]_{i,k}\end{aligned}$$

Σ_{int} is unbiased for the null covariance regardless of the genetic architecture, because the slope (Σ_g) absorbs **all** LD-mediated genetic signal. As a result, the estimator Σ_{int} is approximately unbiased estimator of Σ_K , to the extent $\sigma_k^2 \approx 1$ holds.

Estimating the genetic term in practice. Symmetrically, the slope

$$\hat{\alpha}_{i,k} = \frac{\widehat{\text{Cov}}(\ell_j, z_{j,i} z_{j,k})}{\widehat{\text{Var}}(\ell_j)}$$

derived above via the same normal equations is the OLS estimator of $[\Sigma_g]_{i,k}$, just as $\hat{c}_{i,k}$ is the OLS

estimator of $[\Sigma_{int}]_{i,k}$. Since the regression is well-specified, $\hat{\alpha}_{i,k}$ is likewise unbiased: $\mathbb{E}[\hat{\alpha}_{i,k}] = [\Sigma_g]_{i,k}$. A single OLS fit of $z_{j,i}z_{j,k}$ on ℓ_j across all M SNPs therefore recovers, as its slope and intercept respectively, unbiased estimators of both matrices. Specifically, the estimator $[\hat{c}_{i,k}]_{i,k}$ is used in the omnibus statistic computation $S = \mathbf{z}^\top \hat{\Sigma}_K^{-1} \mathbf{z}$.

□

10.2 Detailed derivation of JASS-inspired null covariance matrix

Low-signal estimator. For each pair of phenotypes (i, j) , the correlation is estimated using only SNPs that are null-like for *both* traits simultaneously:

$$\hat{\Sigma}_{ij} = \text{corr}(z_i z_j : |z_i| < 1.96 \text{ and } |z_j| < 1.96).$$

The threshold $|z| < 1.96$ corresponds to p-value $p > 0.05$, so the retained SNPs are unlikely to carry genuine genetic signal for either trait. Restricting to this low-signal subset ensures that the estimated covariance reflects the null correlation structure which arises from sample overlap and phenotypic correlation alone, not shared genetic effects. By construction, a minimum of 1,000 SNPs is set per pair and if this threshold is not met, the off-diagonal entry defaults to zero. Diagonal entries are fixed at 1 by construction which is an acceptable value as an estimator. To be more rigorous:

Z-score decomposition. Note the z-score $z_{n,i} = \frac{\hat{\beta}_{n,i}}{\text{SE}(\hat{\beta}_{n,i})}$. The estimator $\hat{\beta}_{n,i}$ is unbiased, meaning it centers on the true effect $\mu_{n,i}$: $\hat{\beta}_{n,i} \sim \mathcal{N}(\mu_{n,i}, \text{SE}^2)$. So when we divide by SE: $z_{n,i} = \frac{\hat{\beta}_{n,i}}{\text{SE}} \sim \mathcal{N}(\frac{\mu_{n,i}}{\text{SE}}, 1)$ which decomposes as:

$$z_{n,i} = \underbrace{\frac{\mu_{n,i}}{\text{SE}}}_{\text{signal}} + \underbrace{\varepsilon_{n,i}}_{\varepsilon \sim \mathcal{N}(0,1)}$$

So $\mu_{n,i}$ is the true genetic effect size. $\text{SE}(\hat{\beta})$ is the precision of the estimator, driven by sample size and allele frequency. $\frac{\mu_{n,i}}{\text{SE}}$ together form the signal-to-noise ratio. Under the null: $\mu_{n,i} = 0$, so $z_{n,i} = \varepsilon_{n,i} \sim \mathcal{N}(0, 1)$: pure noise, centered at zero. Under the alternative: $\mu_{n,i} \neq 0$. This means that the larger the true effect relative to its SE, the further the z-score drifts from zero, and the more likely $|z| > 1.96$.

Therefore, for any SNP j and phenotype k , the z-score decomposes as:

$$z_{j,k} = \underbrace{\frac{\mu_{j,k}}{\text{SE}}}_{\text{true genetic signal}} + \underbrace{\varepsilon_{j,k}}_{\text{noise}}$$

Including causal SNPs contaminates the covariance. The covariance between two phenotypes k_1 and k_2 estimated across SNPs is:

$$\text{Cov}(z_{k_1}, z_{k_2}) = \underbrace{\text{Cov}(\varepsilon_{k_1}, \varepsilon_{k_2})}_{\text{null structure}} + \underbrace{\mu_{k_1} \cdot \mu_{k_2}}_{\text{genetic contamination}}$$

The second term is non-zero whenever a SNP has a true effect on **both** traits (pleiotropy). Including such SNPs inflates the covariance estimate beyond the null structure, because shared genetic signal mimics correlation even in the absence of sample overlap.

Remove this contamination. Most SNPs genome-wide are truly null (the genome is sparse). Causal SNPs are the minority but tend to produce large $|z|$ values. By requiring $|z_{k_1}| < 1.96$ **and** $|z_{k_2}| < 1.96$ simultaneously, we approximately select SNPs where $\mu_{k_1} \approx 0$ and $\mu_{k_2} \approx 0$ for both traits. The contaminating term $\mu_{k_1} \cdot \mu_{k_2} \approx 0$, and what remains is:

$$\text{Cov}(z_{k_1}, z_{k_2}) \big|_{\text{null SNPs}} \approx \text{Cov}(\varepsilon_{k_1}, \varepsilon_{k_2})$$

For a null SNP, both z-scores are pure noise drawn from the same GWAS samples. The noise is correlated between phenotypes k_1 and k_2 for two reasons:

1. **Sample overlap** — if the same individuals contributed to both GWAS, any random fluctuation in genotype group composition affects both z_{j,k_1} and z_{j,k_2} simultaneously.
2. **Phenotypic correlation** — if traits k_1 and k_2 are correlated in the sample (e.g. two brain volume measures that co-vary), a random genotype imbalance moves both z-scores in the same direction.

Neither of these is a genetic effect. They are properties of the study design and the phenotypic relationships. This is precisely what the null covariance structure Σ must capture, so that Σ^{-1} can correctly whiten the z-score vector in the joint test.

Summary

The $|z| < 1.96$ filter isolates the null SNPs whose covariance reflects only the study design, not the genetic architecture, giving an uncontaminated estimate of Σ .

Null SNPs have $\mu \approx 0$, so their z-scores are pure noise. The covariance of pure noise between two phenotypes measured in overlapping samples reflects only sample overlap and phenotypic correlation, NOT genetics. Causal SNPs have $\mu \neq 0$ and inject their shared genetic architecture into the covariance estimate, biasing it. The $|z| < 1.96$ filter is a practical way to approximately isolate the null SNPs.

.....

10.3 Claim: Rescaling Transformation Preserves Tail Probabilities

The rescaling step relies on two special functions from `scipy.special`.

Proof. **The χ_k^2 distribution.** A random variable $X \sim \chi_k^2$ has probability density function

$$f(x; k) = \frac{x^{(k/2)-1} e^{-x/2}}{2^{k/2} \Gamma(k/2)}, \quad x \in (0, \infty),$$

where $\Gamma(a) = \int_0^\infty t^{a-1} e^{-t} dt$ is the gamma function; otherwise 0.

chdtrc(k, x): chi-squared survival function. The tail probability $1 - CDF = P(\chi_k^2 > x)$ is obtained by integrating f from x to ∞ w.r.t t . Substituting $u = t/2$:

$$P(\chi_k^2 > x) = \int_x^\infty f(t; k) dt = \frac{1}{\Gamma(k/2)} \int_{x/2}^\infty u^{k/2-1} e^{-u} du = \frac{\Gamma(k/2, x/2)}{\Gamma(k/2)} \equiv Q\left(\frac{k}{2}, \frac{x}{2}\right),$$

where $\Gamma(a, z) = \int_z^\infty t^{a-1} e^{-t} dt$ is the upper incomplete gamma function and $Q(a, z) = \Gamma(a, z)/\Gamma(a)$ is its regularised form. Hence

$$\text{chdtrc}(k, x) = Q\left(\frac{k}{2}, \frac{x}{2}\right) = 1 - P\left(\frac{k}{2}, \frac{x}{2}\right)$$

where $P\left(\frac{k}{2}, \frac{x}{2}\right)$ is the CDF for gamma random variables with shape parameter $\frac{k}{2}$ and scale parameter 2. Note that $Q(a, z) + P(a, z) = 1$ by definition (the upper and lower tails sum to 1).

chdtri(k, p): chi-squared inverse survival function. Given a probability $p \in (0, 1)$, **chdtri** returns the value x^* satisfying $P(\chi_k^2 > x^*) = p$, i.e. it solves

$$Q\left(\frac{k}{2}, \frac{x^*}{2}\right) = 1 - P\left(\frac{k}{2}, \frac{x^*}{2}\right) = p \implies x^* = 2 P^{-1}\left(\frac{k}{2}, 1 - p\right),$$

where $P^{-1}(a, \cdot)$ is the inverse CDF (quantile function) of Gamma $\left(\frac{k}{2}, 1\right)$ with respect to its second argument, i.e. $P^{-1}(a, q)$ returns the value z such that $P(a, z) = q$. This is the inverse survival function (ISF) of χ_k^2 :

$$\text{chdtri}(k, p) = F_{\chi_k^2}^{-1}(1 - p).$$

In the rescaling step, **chdtrc** is called with the subset's k , and **chdtri** is called with $k = 1$ to map onto the χ_1^2 scale as shown below:

Application to the rescaling chain. Given the omnibus statistic $S_k = \|\mathbf{Y}\|^2 \sim \chi_k^2$, the two functions implement the chain

$$S_k \xrightarrow{\text{chdtrc}(k, \cdot)} p = P(\chi_k^2 > S_k) \xrightarrow{\text{chdtri}(1, \cdot)} S_1 = F_{\chi_1^2}^{-1}(1 - p),$$

so that $P(\chi_1^2 > S_1) = p = P(\chi_k^2 > S_k)$: the same tail probability, now expressed on a common χ_1^2 scale regardless of k .

□

.....

10.4 Matrix Inversion via Cholesky factorization

Rather than forming Σ_k^{-1} explicitly at $O(k^3)$ cost, the submatrix is Cholesky factorized as $\Sigma_k = \mathbf{L}\mathbf{L}^\top$ where $\mathbf{L}$ is a lower triangular matrix, and the triangular system $\mathbf{L}\mathbf{Y}_j = \mathbf{z}_j$ is solved for $\mathbf{Y}_j = \mathbf{L}^{-1}\mathbf{z}_j$ at $O(k^2)$ cost. Since $\Sigma_k = \mathbf{L}\mathbf{L}^\top$:

$$\mathbf{Y}_j \sim \mathcal{N}(\mathbf{0}, \mathbf{L}^{-1}\Sigma_k(\mathbf{L}^{-1})^\top) = \mathcal{N}(\mathbf{0}, \mathbf{I}_k),$$

so the k components of $\mathbf{Y}_j$ are i.i.d. $\mathcal{N}(0, 1)$ under H_0 , and the statistic reduces to a simple sum of squares:

$$S_k(j) = \mathbf{z}_j^\top \Sigma_k^{-1} \mathbf{z}_j = \mathbf{z}_j^\top (\mathbf{L}\mathbf{L}^\top)^{-1} \mathbf{z}_j = \mathbf{z}_j^\top (\mathbf{L}^\top)^{-1} \mathbf{L}^{-1} \mathbf{z}_j = (\mathbf{L}^{-1} \mathbf{z}_j)^\top (\mathbf{L}^{-1} \mathbf{z}_j) = \|\mathbf{Y}_j\|^2 \sim \chi_k^2,$$

using $(AB)^{-1} = B^{-1}A^{-1}$ and $(A^{-1})^T = (A^T)^{-1}$ given A and B are invertible.

Glossary of Key Terms

Allele An allele is a variant of the sequence of nucleotides at a particular location on the genome. Alleles can differ at a single position through SNPs. The *effect allele* is the allele whose count is used as the predictor in regression or the one being studied for its association with the phenotype. For example, at a SNP there are two bases **A** and **T**, and set **A** as the effect allele. We can then count **how many copies of A** a person carries:

Genotype	Chromosomes	Copies of “A”	G_i value
TT	T + T	0 copies	0
AT	A + T	1 copy	1
AA	A + A	2 copies	2

where G_i is the genotype of the i -th SNP. Stacking G_i across all N individuals and M SNPs gives the genotype matrix $\mathbf{X} \in \mathbb{R}^{N \times M}$, where $X_{i,j}$ is the copy count of the effect allele for individual i at SNP j .

BLAS (Basic Linear Algebra Subprograms) A library of highly optimised linear algebra routines written in compiled C and Fortran, called internally by `scipy.linalg`. A single BLAS call to `solve_triangular` solves all SNPs simultaneously in one batched operation rather than looping over individual SNPs in Python.

Bonferroni correction A multiple-testing adjustment that multiplies each p-value by the number of tests performed, controlling the probability of any false positive across all tests (Dunn, 1961). In this report it is applied to correct for 10,000 random subset tests per SNP.

Causal effect The true, isolated biological effect of a genetic variant on a phenotype, defined as the change in phenotype that would result from directly altering that variant while holding all others constant. Causal effects are unobserved; GWAS produces marginal estimates that are confounded by LD with neighboring variants.

Cholesky decomposition A factorization of a symmetric positive definite matrix Σ into a lower triangular matrix L such that $\Sigma = LL^\top$. It allows the system $\Sigma^{-1}\mathbf{z}$ to be solved efficiently via forward substitution without explicitly forming the inverse, and is numerically more stable than direct inversion. (See Section 10.4 for more details)

DNA Deoxyribonucleic acid, the molecule that encodes an organism’s genetic information. It consists of two complementary strands wound into a double helix, each strand a sequence of nucleotides (Adenine, Thymine, Guanine, Cytosine); the pairing of nucleotides across the two strands is what is meant by a *base pair*. Genetic variation between individuals – including the SNPs analysed in this report – arises from differences in this sequence.

float32, float64 Two levels of floating-point precision. Float32 (single precision) uses 4 bytes per number and represents approximately 7 significant decimal digits; Float64 (double precision) uses 8 bytes and approximately 15–16 digits. The z-score matrix and linear algebra are carried out in Float32 to halve memory usage, while chi-squared survival functions are evaluated in Float64 to avoid underflow for strongly associated SNPs.

Genome The complete set of DNA in an organism, containing all of its genetic information. In

humans, the genome comprises approximately 3 billion base pairs organised across 23 pairs of chromosomes. This report analyses genetic variation across the human genome at approximately 9.6 million positions.

Genome-wide significance The conventional threshold for declaring a SNP–phenotype association statistically significant after accounting for the large number of SNPs tested simultaneously across the genome. Set at p-value $p < 5 \times 10^{-8}$, corresponding to a chi-squared statistic of approximately 29.7 on one degree of freedom $\approx |z| > 5.45$.

Genotype The specific combination of alleles an individual carries at given genomic positions. For a SNP with two possible alleles, an individual’s genotype is encoded as 0, 1, or 2 copies of the effect allele. GWAS regresses phenotypes on genotypes across all individuals in the cohort.

GWAS summary statistics The per-SNP outputs of a GWAS: for each SNP, a regression of the phenotype on genotype across individuals yields an estimated effect size $\hat{\beta}$ and its standard error $SE(\hat{\beta})$. This report operates entirely on these derived statistics rather than on individual-level genotype data (a full matrix $\mathbf{X}$ with one row per person, one column per SNP, with the 0/1/2 copy counts i.e., know exactly which person has which genotype).

Kernel Density Estimation (KDE) A non-parametric method that estimates the probability density function of a random variable by placing a smooth kernel at each observed value and summing them. Used here to visualise the distribution of chi-squared statistics across SNPs or replicates.

LD correlation (r^2) A measure of the statistical dependence between two SNPs, defined as the squared Pearson correlation between their genotype vectors across individuals: $r_{j,k}^2 = \text{corr}(\mathbf{x}_j, \mathbf{x}_k)^2$. A value of $r^2 = 1$ means the two SNPs are in perfect LD and carry identical information; $r^2 = 0$ means they are completely independent. LD correlation decays with physical distance between SNPs on the chromosome.

LD score A summary measure for a given SNP that quantifies how much total LD it has with all other SNPs in the genome. Formally, the LD score of SNP j is defined as $\ell_j = \sum_{l \neq j} r_{j,l}^2$, the sum of squared correlations between SNP j and every other SNP l . SNPs in regions of high LD have large LD scores; SNPs in low-LD regions have small ones. LD scores are used in LDSC regression to separate genetic signal from noise.

LD Score Regression (LDSC) A method that regresses the product of z-scores $z_i z_j$ across SNPs on LD scores. The intercept of this regression isolates the null covariance arising from sample overlap and phenotypic correlation, removing contamination from shared genetic effects [6]. The **LDSC intercept matrix** is used throughout this report as the null covariance matrix Σ_K .

Linkage disequilibrium (LD) The statistical correlation between nearby genetic variants arising from shared ancestry. SNPs close together on the genome tend to be inherited together and therefore have correlated genotypes in the population. LD means that marginal GWAS estimates for nearby SNPs are correlated even if their true causal effects are independent.

Memory-mapping A technique that maps a file on disk directly into the process’s virtual address space, so the operating system loads only the pages actually accessed rather than the entire file. This provides near-instant loading and a RAM footprint proportional to the data accessed, not the file size. Used throughout for large matrices such as `stat_matrix.dat`.

Non-parametric method A statistical method that does not assume the data follow a specific parametric distribution. Instead, it estimates quantities directly from the observed data. Kernel Density Estimation is a non-parametric method: it estimates the probability density of a variable without assuming it is, for example, normally distributed.

Omnibus test statistic A single summary statistic for the joint association of a SNP with k phenotypes, defined as the Mahalanobis distance $S = \mathbf{z}^\top \Sigma_k^{-1} \mathbf{z}$, where $\mathbf{z}$ is the vector of z-scores and Σ_k is the $k \times k$ null covariance matrix. Under the null hypothesis, $S \sim \chi_k^2$.

Phenotype Any observable or measurable characteristic of an individual, such as height, disease status, or a brain MRI measurement. In this report, 1,010 brain MRI phenotypes derived from UK Biobank participants are analyzed.

Pleiotropy The phenomenon in which a single genetic variant (SNP) has a true causal effect on more than one phenotype simultaneously. Pleiotropic SNPs are of particular interest to multitrait analysis, since their effect may be too weak to reach significance in any single univariate test yet be jointly supported once the correlated phenotypes it affects are considered together.

Positive definite matrix A symmetric matrix Σ is positive definite if $\mathbf{x}^\top \Sigma \mathbf{x} > 0$ for all non-zero vectors $\mathbf{x}$, equivalently if all its eigenvalues are strictly positive. Positive definiteness is required for the Cholesky decomposition to be valid.

p-value The probability, under the null hypothesis, of observing a test statistic at least as extreme as the one computed. A smaller p-value indicates stronger evidence against the null. The conventional genome-wide significance threshold is $p < 5 \times 10^{-8}$.

Q–Q plot (quantile–quantile plot) A diagnostic plot that compares two sets of values by plotting one against the other, with a $y = x$ reference line; points lying on the line indicate agreement, while systematic departures reveal where one set is consistently larger or smaller. In a strict Q–Q plot the two sets are the ordered quantiles of two distributions.

Single Nucleotide Polymorphism (SNP) A single position in the genome where individuals in the population carry different nucleotide bases (A, T, C, G). Each SNP is represented as an effect-allele count (0, 1, or 2 copies) per individual. This report analyses approximately 9.6 million SNPs.

UK Biobank (UKB) A large-scale biomedical database and research resource containing genetic and health information from approximately 500,000 UK participants. This report uses GWAS summary statistics derived from a UKB cohort of 32,947 individuals across 1,010 brain MRI phenotypes.

Variant Any position in the genome where the DNA sequence differs between individuals in the population. SNPs are the most common class of variant, involving a change at a single base pair (fundamental unit of DNA). The terms *variant* and *SNP* are used interchangeably in this report.

z-score The standardized measure of association strength, defined as $z = \hat{\beta}/\text{SE}(\hat{\beta})$ where SE stands for standard error. Under the null hypothesis of no genetic effect, $z \sim \mathcal{N}(0, 1)$. Squaring a z-score gives a chi-squared statistic with one degree of freedom, and $|z| > 5.45$ corresponds to genome-wide significance.